

Book 2

Assignments in SENIOR MATHS

Kevin Dunne



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B o o k 2

Assignments in
**SENIOR
MATHS**

Kevin Dunne



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This book has been written to provide comprehensive coverage of selected topics from the senior Mathematics syllabus. Book 2 concentrates on sequences and series, quadratic equations, the parabola and locus.

Each worksheet has been carefully graded, with the more difficult questions towards the end of each set. Every exercise has an example done to show the most common methods of working. The lessons should be able to be completed by students with minimal help from the teacher. Each unit should have sufficient questions to occupy students for at least a 40-minute lesson.

I would like to thank my wife, Rikki, for her invaluable assistance in the preparation of this book.

Kevin Dunne

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Recognition of arithmetic sequences (progressions)

To show that a sequence is arithmetic you need to show that:

$$T_2 - T_1 = T_3 - T_2 = d [= T_4 - T_3] \text{ etc}$$

Example: Show that 4, 9, 14, ... is arithmetic.

Solution: $9 - 4 = 5$ and $14 - 9 = 5$

thus the sequence is arithmetic with the common difference of 5.

Determine whether the following sets of numbers are arithmetic sequences. If they are, write down the next two terms.

1. 9, 10, 11, __, __

2. 8, 15, 21, __, __

3. 10, 20, 30, __, __

4. -4, -2, 0, __, __

5. 4, 8, 16, __, __

6. -1, 3, 7, __, __

7. 8, 16, 24, __, __

8. 6, 10, 14, __, __

9. 3, 12, 21, __, __

10. 6, 12, 17, __, __

11. -4, 0, 4, __, __

12. -6, -4, 0, __, __

13. -5, -3, -1, __, __

14. 20, 40, 80, __, __

15. 101, 112, 123, __, __

16. 9, 7, 5, __, __

17. 8, 4, 1, __, __

18. 7, 3, -1, __, __

19. -10, -12, -14, __, __

20. -20, -16, -12, __, __

21. $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, _, _$

22. $\frac{1}{10}, \frac{1}{5}, \frac{3}{10}, _, _$

23. $\frac{1}{6}, \frac{1}{4}, \frac{1}{3}, _, _$

24. $\frac{1}{5}, \frac{1}{10}, \frac{1}{15}, _, _$

25. $2\frac{1}{2}, 3\frac{1}{4}, 4, _, _$

26. $1\frac{1}{2}, 1\frac{1}{5}, \frac{9}{10}, _, _$

27. 0.05, 0.1, 0.15, __, __

28. 3.2, 4.0, 4.8, __, __

29. 8.4, 8.0, 7.6, __, __

30. 9.4, 8.9, 8.5, __, __

31. $a, 2a, 3a, _, _$

32. $a, 2a + 3, 3a + 6, _, _$

33. $a, 3a, 6a, _, _$

34. $\sqrt{3}, \sqrt{6}, \sqrt{9}, _, _$

35. $\sqrt{3}, 4\sqrt{3}, 7\sqrt{3}, _, _$

36. $\sqrt{5}, 5, 5\sqrt{5}, _, _$

37. $\sqrt{6} + 4, 2\sqrt{6} + 6, 3\sqrt{6} + 8, _, _$

38. $3, 3 + \sqrt{3}, 3 + \sqrt{12}, _, _$

39. $\sqrt{2}, \sqrt{8}, \sqrt{18}, _, _$

40. $\sqrt{5} + 1, \sqrt{7} + 2, \sqrt{9} + 3, _, _$

Finding terms of arithmetic sequences

To find a particular term of an arithmetic sequence use: $T_n = a + (n - 1)d$
 where: a is the first term, d is the common difference and n is the number of terms.

Example: Determine the fifteenth term of the sequence 0, 6, 12, ____, ____

Solution: $a = 0$, $d = 6$ and $n = 15$

$$T_n = a + (n - 1)d$$

$$\begin{aligned} T_{15} &= 0 + 14 \times 6 \\ &= 84 \end{aligned}$$

$\therefore$ fifteenth term is 84

For each sequence determine the term indicated.

1. 6, 8, 10, ... T_7

2. 9, 13, 17, ... T_6

3. 8, 12, 16, ... T_{10}

4. 0, 3, 6, ... T_{15}

5. 1, 6, 11, ... T_{20}

6. -4, 0, 4, ... T_{17}

7. 9, 15, 21, ... T_{20}

8. -4, -1, 2, ... T_{10}

9. 0, 8, 16, ... T_{30}

10. 3, 9, 15, ... T_{37}

11. 6, 4, 2, ... T_{20}

12. 16, 8, 0, ... T_{14}

13. -6, -10, -14, ... T_{20}

14. -6, -4, -2, ... T_{30}

15. 20, 28, 36, ... T_{40}

16. 9.3, 9.6, 9.9, ... T_{15}

17. 0.3, 1.4, 2.5, ... T_{21}

18. 4.2, 3.4, 2.6, ... T_{16}

19. -1.3, -2.6, -3.9, ... T_{40}

20. 2.8, 2.0, 1.2, ... T_{25}

21. $\frac{1}{4}$, $1\frac{1}{2}$, $2\frac{3}{4}$, ... T_{24}

22. $\frac{1}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, ... T_{27}

Write down the missing terms in the following arithmetic sequences:

23. ____, ____, 8, 12, 16

24. ____, ____, 9, 14, 19

25. ____, 6, ____, 16, 21

26. ____, 0, 6, 12, ____

27. ____, 12, 9, 6, ____

28. ____, ____, 8, 17, 26

29. 6, 1, ____, ____, -14

30. ____, ____, $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$

31. ____, ____, 7.4, 8.0, 8.6

32. ____, ____, $\frac{1}{5}$, $\frac{3}{5}$, 1

Recognising terms of arithmetic sequences

To show that a number is a term of an arithmetic sequence use: $T_n = a + (n - 1)d$

Example: Show that 72 is a term of the sequence $-3, 0, 3, \dots$

Solution: $a = -3, d = 3$ and $T_n = 72$

$$72 = -3 + (n - 1) \times 3$$

$$72 = -3 + 3n - 3$$

$$78 = 3n$$

$$\therefore n = 26$$

As 26 is a positive integer, 72 is a term of the sequence.

In each question the number in brackets is a term of the sequence. Determine which term it is.

1. $6, 9, 12, \dots$ (78)

2. $9, 15, 21, \dots$ (183)

3. $-3, 5, 13, \dots$ (309)

4. $8, 5, 2, \dots$ (-70)

5. $19, 15, 11, \dots$ (-161)

6. $15, 8, 1, \dots$ (-230)

7. $6, 11, 16, \dots$ (156)

8. $4, 9, 14, \dots$ (134)

9. $6, 1, -4, \dots$ (-94)

10. $14, 21, 28, \dots$ (322)

11. $9, 17, 25, \dots$ (345)

12. $0, -3, -6, \dots$ (-141)

13. $20, 31, 42, \dots$ (218)

14. $-8, -2, 4, \dots$ (424)

15. $24, 20, 16, \dots$ (-424)

16. $13, 20, 27, \dots$ (118)

17. $12, 6, 0, \dots$ (-264)

18. $2, 20, 38, \dots$ (506)

19. $16, 8, 0, \dots$ (-88)

20. $5, 18, 31, \dots$ (421)

21. $0.4, 1.1, 1.8, \dots$ (18.6)

22. $3.6, 5, 6.4, \dots$ (59.6)

23. $7.2, 7, 6.8, \dots$ (-16.6)

24. $-0.3, -0.7, -1.1, \dots$ (-11.1)

25. $\frac{3}{4}, 1\frac{1}{2}, 2\frac{1}{4}, \dots$ ($33\frac{3}{4}$)

26. $\frac{5}{7}, \frac{3}{7}, \frac{1}{7}, \dots$ (-7)

27. $\frac{1}{5}, \frac{3}{10}, \frac{2}{5}, \dots$ (3)

28. $2\frac{1}{5}, 3\frac{1}{3}, 4\frac{7}{15}, \dots$ ($52\frac{1}{15}$)

29. $\sqrt{5}, \sqrt{20}, \sqrt{45}, \dots$ ($\sqrt{3645}$)

30. $7x, 12x, 17x, \dots$ ($177x$)

Finding the arithmetic sequence given two terms

Example: The fifth term of an arithmetic sequence is 23 and the sixteenth term is 78.

Determine the first term and the common difference.

Solution: You will need to use the formula $T_n = a + (n - 1)d$, and solve simultaneous equations

$$a + 4d = 23 \quad (1)$$

$$a + 15d = 78 \quad (2)$$

$$(2) - (1)$$

$$11d = 55$$

$$d = 5$$

Substitute $d = 5$ into (1)

$$a = 3$$

$\therefore$ the first term is 3 and the common difference is 5.

Two terms of an arithmetic sequence are given, determine the first term and the common difference.

- | | |
|----------------------------|---------------------------|
| 1. $T_5 = 18,$ | $T_{16} = 51$ |
| 2. $T_4 = 4,$ | $T_{11} = 18$ |
| 3. $T_4 = 0,$ | $T_{12} = 16$ |
| 4. $T_6 = 36,$ | $T_{11} = 71$ |
| 5. $T_1 = 9,$ | $T_6 = 44$ |
| 6. $T_3 = 4,$ | $T_{12} = 49$ |
| 7. $T_2 = -6,$ | $T_{10} = -30$ |
| 8. $T_9 = 14,$ | $T_{21} = 20$ |
| 9. $T_7 = \sqrt{3} + 18,$ | $T_{20} = \sqrt{3} + 57$ |
| 10. $T_5 = 6 + \sqrt{48},$ | $T_{10} = 6 + \sqrt{243}$ |

For each sequence two terms are given, determine the first two terms.

- | | |
|----------------------------|--------------------------|
| 11. $T_{17} = 48,$ | $T_{20} = 120$ |
| 12. $T_5 = -0.7,$ | $T_{15} = 1.3$ |
| 13. $T_4 = -27,$ | $T_8 = -57$ |
| 14. $T_6 = -18,$ | $T_{17} = -62$ |
| 15. $T_4 = 4.4,$ | $T_{10} = 12.2$ |
| 16. $T_3 = 1\frac{3}{4},$ | $T_{11} = 17\frac{3}{4}$ |
| 17. $T_7 = -2\frac{1}{5},$ | $T_{12} = -4\frac{1}{5}$ |
| 18. $T_4 = 1\frac{1}{5},$ | $T_9 = 2\frac{13}{15}$ |
| 19. $T_3 = -2\frac{2}{5},$ | $T_7 = -2\frac{1}{5}$ |
| 20. $T_5 = -7\sqrt{3},$ | $T_{15} = -27\sqrt{3}$ |

Finding a given term of an arithmetic sequence

Example: Determine whether 116 is a term of the sequence 6, 10, 14, ...

Solution: $T_n = a + (n - 1)d$

$$116 = 6 + (n - 1)4$$

$$116 = 6 + 4n - 4$$

$$114 = 4n$$

$$n = 28\frac{1}{2}$$

As n is not a positive integer, 116 is NOT a term of the sequence.

In each question find whether the number in parentheses is a term of the given arithmetic sequence.

1. 3, 8, 13, ... (188)

11. 7, 12, 17, ... (177)

2. 4, 11, 18, ... (214)

12. 1, -2, -5, ... (-83)

3. 6, 9, 12, ... (142)

13. -9, -5, -1, ... (191)

4. 5, 13, 21, ... (277)

14. -1, 3, 7, ... (171)

5. -4, 1, 6, ... (116)

15. 0.2, 0.9, 1.6, ... (19.4)

6. -19, -16, -13, ... (66)

16. -0.4, -1.0, -1.6, ... (-20.6)

7. 6, 2, -2, ... (-182)

17. $\frac{3}{5}, \frac{4}{5}, 1, \dots (8\frac{2}{5})$

8. -8, -14, -20, ... (-170)

18. $-\frac{2}{7}, -\frac{1}{7}, 0, \dots (-7\frac{4}{7})$

9. 0.3, 0.9, 1.5, ... (27.2)

19. $-\frac{4}{9}, -\frac{7}{9}, -1\frac{1}{9}, \dots (-13\frac{1}{9})$

10. 0.5, 1.2, 1.9, ... (14.7)

20. $\frac{1}{5}, \frac{9}{20}, \frac{7}{10}, \dots (8\frac{9}{20})$

21. Determine the largest term of the sequence 8, 14, 20, ... that is smaller than 192.

22. Determine the largest term of the sequence -3, -1, 1, ... that is smaller than 118.

23. Determine the smallest term of the sequence 7, 12, 17, ... that is larger than 2067.

24. Determine the term of the sequence 3, 9, 15, ... that is closest to 224.

25. Determine the term of the sequence -5, -12, -19, ... that is closest to -275.

The sum of an arithmetic series (1)

To determine the sum of an arithmetic series, use the formula: $S_n = \frac{n}{2} [2a + (n-1)d]$, where n is the number of terms, a is the first term and d is the common difference.

Example: Find the sum of the first 21 terms of the series $-3 + 1 + 5 \dots$

Solution: $a = -3, d = 4, n = 21$

$$S_{21} = \frac{21}{2} [2 \times -3 + (21-1) \times 4]$$

$$= \frac{21}{2} \times 74$$

$$= 777$$

$\therefore$ the total of 21 terms is 777

Find the sum of each arithmetic series to the number of terms indicated.

1. $6 + 11 + 16 + \dots$ (20 terms)

2. $9 + 12 + 15 + \dots$ (15 terms)

3. $0 + 9 + 18 + \dots$ (17 terms)

4. $3 + 7 + 11 + \dots$ (16 terms)

5. $-4 - 1 + 2 + \dots$ (21 terms)

6. $-9 - 4 + 1 + \dots$ (30 terms)

7. $-10 - 4 + 2 + \dots$ (15 terms)

8. $6 + 4 + 2 + \dots$ (30 terms)

9. $5 + 1 - 3 - \dots$ (14 terms)

10. $16 + 11 + 6 + \dots$ (19 terms)

11. $6 + 0 - 6 - \dots$ (12 terms)

12. $-24 - 22 - 20 - \dots$ (13 terms)

13. $17 + 20 + 23 + \dots$ (10 terms)

14. $-6 - 1 + 4 + \dots$ (18 terms)

15. $-30 - 27 - 24 - \dots$ (12 terms)

16. $50 + 42 + 34 + \dots$ (21 terms)

17. $8\frac{1}{2} + 10 + 11\frac{1}{2} + \dots$ (25 terms)

18. $1\frac{1}{2} + 1 + \frac{1}{2} + \dots$ (17 terms)

19. $\frac{3}{4} + 1\frac{1}{2} + 2\frac{1}{4} + \dots$ (22 terms)

20. $3 + 2\frac{3}{5} + 2\frac{1}{5} + \dots$ (28 terms)

21. The sum of an arithmetic series is 3100, when 31 terms are added together. The common difference is 6. Determine the first term.

22. The sum of an arithmetic series is 1189, when 29 terms are added together. The first term is -1 . Determine the common difference.

23. How many terms of the series $2 + 5 + 8 + \dots$ need to be added together to give a total of 345?

24. How many terms of the series $-9 - 5 - 1 - \dots$ need to be added together to give a total of 391?

25. How many terms of the series $-6 - 4 - 2 - \dots$ need to be added together to give a total of 260?

The sum of an arithmetic series (2)

To determine the sum of an arithmetic series when you are given the first term, a , the last term, l , and the number of terms, n , use the formula: $S_n = \frac{n}{2}(a + l)$

Example: Find the sum of the series $8 + 10 + 12 + \dots + 98$

Note: you first need to find n using: $T_n = a + (n - 1)d$

$$98 = 8 + (n - 1) \times 2$$

$$98 = 8 + 2n - 2$$

$$2n = 92$$

$$n = 46$$

$$\text{Now, } a = 8, l = 98, n = 46$$

$$S_n = \frac{46}{2}(8 + 98)$$

$$= 23 \times 106$$

$$= 2438$$

$\therefore$ The sum of the series is 2438.

Determine the sum of the following series.

1. $6 + 9 + 12 + \dots + 93$ (ie 32 terms)

2. $0 + 5 + 10 + \dots + 105$ (ie 22 terms)

3. $8 + 3 - 2 - 7 \dots - 137$

4. $-4 + 0 + 4 + \dots + 88$

5. $0 - 2.5 - 5 - \dots - 60$

6. $6 + 8 + 10 + \dots 66$

7. $7 + 12 + 17 + \dots 102$

8. $4 + 7 + 10 + \dots 55$

9. $-3 - 7 - 11 - \dots - 67$

10. $6.4 + 6.2 + 6.0 + \dots 1.6$

11. $\frac{1}{8} + \frac{3}{8} + \frac{5}{8} + \dots 4\frac{1}{8}$

12. $-\frac{1}{5} - \frac{3}{5} - 1 \dots - 7\frac{4}{5}$

13. $\sqrt{2} + 3\sqrt{2} + 5\sqrt{2} + \dots 79\sqrt{2}$

14. When 42 terms of an arithmetic series are added together, the sum is 4599. If the first term is 7, find the last term.

15. When 19 terms of an arithmetic series are added together, the sum is 646. If the last term is -29 , determine the first three terms of the series.

Sigma notation and arithmetic series problems

The symbol sigma ($\sum$) is used to indicate that the terms of a series are to be added.

Example: Evaluate $\sum_{n=1}^{15} (3n + 1)$

Solution: $\sum_{n=1}^{15} (3n + 1) = (3 \times 1 + 1) + (3 \times 2 + 1) + (3 \times 3 + 1) + \dots + (3 \times 15 + 1)$
 $= 4 + 7 + 10 + \dots + 46$

Note: Be careful to select the correct value of n .

In this example we have an arithmetic series, where $a = 4$, $l = 46$, and $n = 15$.

Use $S_n = \frac{n}{2} (a + l)$

$S_{15} = \frac{15}{2} (4 + 46)$
 $= 375$

Note: You could have used $S_n = \frac{n}{2} [2a + (n - 1)d]$.

Determine the value of:

1. $\sum_{n=1}^{10} (2n)$

5. $\sum_{n=10}^{20} (5n + 2)$

9. $\sum_{n=5}^{25} (6 - n)$

2. $\sum_{n=1}^{14} (2n - 1)$

6. $\sum_{n=12}^{22} (3 - 2n)$

10. $\sum_{n=1}^{21} (\frac{n}{2} + 3)$

3. $\sum_{n=1}^{15} (4n)$

7. $\sum_{n=2}^{18} (4n + 5)$

4. $\sum_{n=1}^{12} (2n - 4)$

8. $\sum_{n=1}^{30} (3n - 5)$

In each of the following questions, establish two equations and solve them simultaneously.

- The sum of the first 5 terms of an arithmetic series is 50. The sum of the first 18 terms is 414. Determine the first three terms.
- The sum of the first 8 terms of an arithmetic series is 152. The sum of the first 12 terms is 324. Determine the first three terms.
- The sum of the first 10 terms of an arithmetic series is 230. The sum of the first 20 terms is 1060. Determine the first three terms.
- The sum of the first 7 terms of an arithmetic series is 35. The sum of the *next* 6 terms is 186. Determine the first three terms.
- The sum of the first 5 terms of an arithmetic series is -15 . The sum of the *next* 5 terms is -165 . Determine the first three terms.

Arithmetic sequences and series review

You may need the following formulae:

$$T_n = a + (n - 1)d \quad S_n = \frac{n}{2}[2a + (n - 1)d] \quad S_n = \frac{n}{2}(a + l) \quad d = T_2 - T_1 = T_3 - T_2 \text{ etc}$$

1. Is the sequence 3, 6, 20, ... arithmetic?
2. Is the sequence -9, -13, -17, ... arithmetic?
3. Find the missing terms of the arithmetic sequence ____, ____, 0, 9, 18, ...
4. Determine the 21st term of the sequence 6, 13, 20, ...
5. Determine the sum of the first 33 terms of the series $3 + 7 + 11 + \dots$
6. Determine the sum of the first 40 terms of the series $-3 - 9 - 15 - \dots$
7. Which term of the sequence 7, 19, 31, ... is 139?
8. Is -79 a term of the sequence 9, 5, 1, ...?
9. Insert 5 terms between 3 and 51 to form an arithmetic sequence.
10. How many terms of the series $-3 + 3 + 9 + \dots$ must be added to give a total of 1080?
11. Find the sum of the series $4 + 9 + 14 + \dots + 379$.
12. Determine the first term of the sequence 101, 95, 89, ... that is negative.
13. Evaluate $\sum_{n=1}^{35} (2n - 4)$.
14. For an arithmetic sequence $T_3 = 18$ and $T_6 = 33$. Determine the first term and the common difference.
15. If a , $3b$ and $4c$ form an arithmetic sequence, write an expression for a , in terms of b and c .
16. The sum of the first 10 terms of an arithmetic series is 265. The sum of the first 20 terms is 1030. Determine the first three terms.
17. The following shapes are made out of matchsticks:



Shape 1



Shape 2



Shape 3

If this pattern continues, how many matches will be needed to make the 61st shape?

18. How many matches will be needed to make the first 61 shapes in Question 17?
19. A truck delivers gravel to a road construction site. The first load is delivered from the quarry to the start of the road, a distance of 3.8 km. It then returns to the quarry to get the second load. This load is dropped 100 m beyond the first load. Each successive load is dropped 100 m further away than the previous load.
How far will the truck travel to deliver the 41st load and return to the quarry?
20. What distance will the truck travel to make 120 such loads?

Recognition of geometric sequences (progressions)

To show that a sequence is geometric you need to show that $T_2 \div T_1 = T_3 \div T_2 = r [= T_4 \div T_3]$ etc

Example: Show that 8, 12, 18, ... is geometric.

Solution: $12 \div 8 = 1\frac{1}{2}$ and $18 \div 12 = 1\frac{1}{2}$

Thus the sequence is geometric with the common ratio of $1\frac{1}{2}$ or $\frac{3}{2}$.

Some of the sets of numbers form geometric sequences, some do not. Determine which are geometric and write down the next two terms.

1. 2, 4, 8, ____, ____,

2. 5, 10, 15, ____, ____,

3. 1, 2, 9, ____, ____,

4. 10, 20, 40, ____, ____,

5. 5, 10, 45, ____, ____,

6. -2, -8, -32, ____, ____,

7. -3, 0, -3, ____, ____,

8. -4, 8, -16, ____, ____,

9. -2, -6, -18, ____, ____,

10. 80, 40, 20, ____, ____,

11. 9, 27, 81, ____, ____,

12. 4, 2, 0, ____, ____,

13. 0.3, 0.6, 1.2, ____, ____,

14. -0.7, -1.3, -2.8, ____, ____,

15. 8, 4, 2, ____, ____,

16. 2, -6, 12, ____, ____,

17. $7, 3\frac{1}{2}, 1\frac{3}{4},$ ____, ____,

18. -5, -10, -20, ____, ____,

19. $-\frac{1}{2}, -\frac{1}{4}, -\frac{1}{8},$ ____, ____,

20. $1\frac{1}{4}, 2\frac{1}{2}, 5,$ ____, ____,

21. $\sqrt{2}, 2, 2\sqrt{2},$ ____, ____,

22. $\sqrt{3}, 2\sqrt{3}, 4\sqrt{3},$ ____, ____,

23. $\frac{1}{a}, \frac{1}{ab}, \frac{1}{ab^2},$ ____, ____,

24. $\frac{1}{a}, \frac{3}{a}, \frac{6}{a},$ ____, ____,

25. $\frac{x}{y}, \frac{ax}{y^2}, \frac{a^2x}{y^3},$ ____, ____,

26. $\frac{1}{\sqrt{3}}, \sqrt{3}, 3,$ ____, ____,

For each of the following geometric sequences, determine the missing terms.

27. ____, 6, 12, 24, ____,

28. -8, -16, ____, -64, ____,

29. ____, 1, 5, 25, ____,

30. ____, 7, $-\frac{7}{3}, \frac{7}{9},$ ____,

31. ____, ____, $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}$

32. 1.2, 3.6, 10.8, ____, ____,

33. -4, ____, -5.76, -6.912, ____,

34. ____, $\frac{2}{3}, \frac{1}{9}, \frac{1}{54},$ ____,

35. ____, ____, -0.172, -0.0344, -0.00688

36. ____, $1\frac{1}{3}, \frac{8}{9}, \frac{16}{27},$ ____,

Finding terms of geometric sequences

To find a particular term of a geometric sequence, use: $T_n = ar^{n-1}$
where a is the first term, r is the common ratio and n is the number of terms.

Example: Determine the eighth term of the sequence 99, 33, 11.

Solution: $a = 99, r = \frac{1}{3}, n = 8$

$$T_8 = 99 \times \left(\frac{1}{3}\right)^7$$

$$= \frac{11}{243}$$

$$\therefore \text{the eighth term is } \frac{11}{243}$$

For each geometric sequence find the term indicated.

- | | |
|--|---|
| 1. 1, 3, 9, ... (T_6) | 9. $\frac{1}{3}, 1\frac{1}{3}, 5\frac{1}{3}, \dots$ (T_8) |
| 2. -6, -12, -24, ... (T_7) | 10. -1, -4, -16, ... (T_7) |
| 3. 2, 10, 50, ... (T_7) | 11. -12, -6, -3, ... (T_8) |
| 4. 4, 8, 16, ... (T_{10}) | 12. 34, 3.4, 0.34, ... (T_9) |
| 5. -1, 2, -4, ... (T_9) | 13. $x, \frac{x}{3}, \frac{x}{9}, \dots$ (T_7) |
| 6. 10, 5, $2\frac{1}{2}$, ... (T_8) | 14. 5x, 15x, 45x, ... (T_8) |
| 7. -9, -3, -1, ... (T_8) | 15. $2x, 2x^2, 2x^3, \dots$ (T_{10}) |
| 8. 0.5, 1, 2, ... (T_9) | 16. $-6b, 3b, -\frac{3b}{2}, \dots$ (T_8) |

-
17. The sixth term of a geometric sequence is 96 and $r = 2$. Determine the first term.
18. The fifth term of a geometric sequence is -162 and $r = 3$. Determine the first term.
19. The sixth term of a geometric sequence is -1458 and $r = -3$. Determine the first term.
20. The seventh term of a geometric sequence is $\frac{2}{729}$ and $a = 2$. Determine the common ratio.
21. The ninth term of a geometric sequence is 32 and $a = \frac{1}{8}$. Determine the common ratio.
22. The ninth term of a geometric sequence is -729 and $r = 3$. Determine the first term.
23. The seventh term of a geometric sequence is $256x^6$ and $r = 2x$. Determine the first term.
24. The seventh term of a geometric sequence is $7\frac{19}{32}$ and $a = \frac{2}{3}$. Determine the common ratio.
25. The eighth term of a geometric sequence is $\frac{2x^6}{y^7}$ and $r = \frac{2}{x}$. Determine the first term.
26. The tenth term of a geometric sequence is 48 and $r = \sqrt{2}$. Determine the first term.

Recognising terms of geometric sequences

To show that a number is a term of a geometric sequence use: $T_n = ar^{n-1}$

Example: Which term of the geometric sequence $\frac{1}{4}, \frac{3}{4}, \frac{9}{4}, \dots$ is $60\frac{3}{4}$?

Solution: $a = \frac{1}{4}, r = \frac{3}{4}, T_n = 60\frac{3}{4}$

$$60\frac{3}{4} = \frac{1}{4} \times 3^{n-1}$$

$$243 = 3^{n-1}$$

$$3^5 = 3^{n-1}$$

$$\therefore 5 = n - 1$$

$$n = 6$$

$60\frac{3}{4}$ is the sixth term of the sequence.

Note: Use your calculator and repeated multiplication by 3 to find $243 = 3^5$.

In each of the following questions, the number in brackets is a term of the geometric sequence. Determine which term it is.

1. 5, 10, 20 ... (2560)

2. 4, 12, 36, ... (26 244)

3. 6, 3, $1\frac{1}{2}$, ... ($\frac{3}{256}$)

4. 0.4, 0.8, 1.6, ... (6553.6)

5. $\frac{1}{5}, \frac{3}{5}, 1\frac{4}{5}$, ... (1312 $\frac{1}{5}$)

6. -1, -4, -16, ... (-4096)

7. 0.2, 0.8, 3.2, ... (13 107.2)

8. 2, -6, 18, ... (-354 294)

9. -2, -1, $-\frac{1}{2}$, ... ($-\frac{1}{2048}$)

10. -100, -20, -4, ... ($-\frac{4}{3125}$)

11. -300, -150, -75, ... ($-\frac{75}{256}$)

12. 3.2, -6.4, 12.8, ... (3276.8)

13. $\sqrt{3}$, 3, $3\sqrt{3}$, ... (243)

14. $\sqrt{3}$, $\sqrt{6}$, $\sqrt{12}$, ... ($\sqrt{3072}$)

15. a , $3a$, $6a$, ... (19 683 a)

16. $\frac{2}{x}$, 2, $2x$, ... ($2x^{12}$)

17. b^{-5} , b^{-3} , b^{-1} , ... (b^{15})

18. m^2 , mx , x^2 , ... ($\frac{x^{15}}{m^{13}}$)

For each of the following geometric sequences, determine whether the number in brackets is a term of the sequence.

19. 3, 6, 12, ... (96)

20. $\frac{1}{2}, \frac{1}{3}, \frac{2}{9}$, ... ($\frac{32}{729}$)

21. $\frac{1}{18}, \frac{1}{3}, 2$, ... (2592)

22. -2, 14, -98, ... (235 298)

23. $\sqrt{2}$, 2, $\sqrt{8}$, ... (2048)

24. 6, 7.8, 10.14, ... (48.52)

Finding the geometric sequence given two terms

Example: The fourth term of a geometric sequence is 48 and the eighth term is 768. Determine the first term and the common ratio.

You will need to use the formula $T_n = ar^{n-1}$ and be able to solve simultaneous equations involving indices.

$$ar^3 = 48 \quad (1)$$

$$ar^7 = 768 \quad (2)$$

$$(2) \div (1)$$

$$\therefore \frac{ar^7}{ar^3} = \frac{768}{48}$$

$$\therefore r^4 = 16$$

$$r = 2$$

Substitute $r = 2$ in (1)

$$a \times 8 = 48$$

$$a = 6$$

Solution is $a = 6, r = 2$

In each question, two terms of a geometric sequence are given. Determine the first term (a) and the common ratio (r) for each question.

1. $T_3 = 16, T_7 = 256$

9. $T_4 = \frac{3}{4}, T_8 = \frac{3}{64}$

2. $T_7 = 1458, T_9 = 13\,122$

10. $T_3 = -2\frac{2}{3}, T_8 = -\frac{256}{729}$

3. $T_4 = 135, T_6 = 1215$

11. $T_4 = \frac{3}{64}, T_8 = \frac{3}{16\,384}$

4. $T_5 = -324, T_{10} = -78\,732$

12. $T_3 = 1.8, T_7 = 145.8$

5. $T_3 = -98, T_7 = -235\,298$

13. $T_4 = -\frac{1}{3}, T_7 = -\frac{1}{81}$

6. $T_3 = -2.8, T_{10} = -358.4$

14. $T_2 = 30, T_7 = -\frac{10}{81}$

7. $T_3 = -20, T_7 = -320$

15. $T_4 = 0.162, T_6 = 0.01458$

8. $T_4 = 1250, T_7 = 156\,250$

16. $T_4 = 67.5, T_7 = 227.8125$

Sum of a geometric series

There are two formulae for determining the sum of a geometric series.

If ' r ' is between -1 and 1 , ie $|r| < 1$, use $S_n = \frac{a(1 - r^n)}{1 - r}$

For all other values of r , ie $|r| > 1$, use $S_n = \frac{a(r^n - 1)}{r - 1}$

The symbol $\sum$ means the 'sum of terms'.

Example: $\sum_{n=1}^9 3 \times 2^{1-n} = 3 \times 2^{1-1} + 3 \times 2^{1-2} + 3 \times 2^{1-3} \dots$

which can be evaluated using the sum of a geometric series formula.

Find the sum of the following geometric series. Where necessary give answers to two decimal places.

1. $1 + 4 + 16 + \dots$ 10 terms

12. $5 + 1 + \frac{1}{5} + \dots$ 8 terms

2. $3 + 9 + 27 + \dots$ 9 terms

13. $-\frac{1}{4} + \frac{1}{2} - 1 + \dots$ 12 terms

3. $2 + 6 + 18 + \dots$ 9 terms

14. $a + ab + ab^2 + \dots$ 10 terms

4. $7 + 14 + 28 + \dots$ 7 terms

15. $\frac{1}{x} + 1 + x + \dots$ 12 terms

5. $-6 - 12 - 24 - \dots$ 8 terms

16. $\sqrt{2} + 2 + \sqrt{8} + \dots$ 9 terms

6. $-4 - 8 - 16 - \dots$ 10 terms

17. $\sum_{n=1}^8 (3 \times 2^{n-1})$

7. $-2 + 4 - 8 + \dots$ 16 terms

18. $\sum_{n=1}^9 (3 \times 4^{1-n})$

8. $16\,384 + 4096 + 1024 + \dots$ 8 terms

19. $\sum_{n=1}^8 (\frac{1}{3} \times (\frac{1}{2})^{n-1})$

9. $\frac{1}{3} + 1 + 3 + \dots$ 11 terms

20. $\sum_{n=1}^{10} (3 \times (0.4)^{n-1})$

10. $-12 - 6 - 3 - \dots$ 9 terms

21. $\sum_{n=1}^9 -2 \times 2^{n-1}$

11. $10 + 12 + 14.4 + \dots$ 7 terms

22. $\sum_{n=3}^9 100 \times (0.2)^{n-3}$

The limiting sum of a geometric series

The limiting sum of a geometric series occurs when an 'infinite' number of terms are added together. For the sum to exist, the common ratio must be between -1 and 1 , ie $|r| < 1$.

The limiting sum formula is: $S_{\infty} = \frac{a}{1-r}$

Some of the following series have a limiting sum. Determine this sum where it exists.

- | | |
|--|--|
| 1. $10 + 5 + 2\frac{1}{2} + \dots$ | 10. $10 - 5 - 0 - \dots$ |
| 2. $3 + 6 + 12 + \dots$ | 11. $16 + 4 + 1 + \dots$ |
| 3. $100 + 50 + 25 + \dots$ | 12. $4 - 2 + 1 - \dots$ |
| 4. $\frac{1}{4} + \frac{1}{2} + 1 + \dots$ | 13. $\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$ |
| 5. $1 + \frac{1}{3} + \frac{1}{9} + \dots$ | 14. $1 - \frac{3}{4} + \frac{9}{16} - \dots$ |
| 6. $1 + 1\frac{1}{2} + 2\frac{1}{4} + \dots$ | 15. $-8 + 4 - 2 + \dots$ |
| 7. $-2 - 4 - 8 - \dots$ | 16. $\frac{3}{2} + \frac{3}{5} + \frac{6}{25} + \dots$ |
| 8. $25 + 5 + 1 + \dots$ | 17. $3 + 1.8 + 1.08 + \dots$ |
| 9. $6 + 4 + 2\frac{2}{3} + \dots$ | 18. $8.4 + 1.68 + 0.336 + \dots$ |

19. A geometric series has a first term of 5 and a common ratio of $\frac{3}{5}$. Determine the limiting sum.
20. Evaluate $\sum_{n=1}^{\infty} (8 \times (\frac{3}{4})^{n-1})$
21. A geometric series has a common ratio of $\frac{1}{4}$ and a limiting sum of $9\frac{1}{3}$. Determine the first term.
22. A geometric series has a first term of 6. If the fourth term is $2\frac{17}{32}$, determine the limiting sum.
23. The lava flow of a volcano after one week is 200 m. Each week the flow increases by $\frac{9}{10}$ of the previous week's length. Determine the maximum length of the lava flow.

Recurring decimals and the limiting sum

A recurring decimal can be expressed as a fraction by using the limiting sum of a geometric series.

$$\text{ie } S_{\infty} = \frac{a}{1-r}$$

Example: Express $0.\dot{1}\dot{4}$ as a fraction.

Solution: $0.\dot{1}\dot{4} = 0.14141414 \dots = \frac{14}{100} + \frac{14}{10\,000} + \frac{14}{1\,000\,000} + \dots$

This is the limiting sum of a geometric series, where $a = \frac{14}{100}$ and $r = \frac{1}{100}$.

$$\begin{aligned} 0.\dot{1}\dot{4} &= \frac{\frac{14}{100}}{1 - \frac{1}{100}} \\ &= \frac{14}{99} \end{aligned}$$

Note: The use of a geometric series is not the only method for converting recurring decimals to fractions.

Convert each of the following recurring decimals to fractions or mixed numbers.

1. $0.\dot{2}$

12. $0.0\dot{8}$

2. $0.\dot{4}$

13. $6.\dot{7}$

3. $0.\dot{2}\dot{1}$

14. $5.\dot{1}\dot{4}$

4. $0.\dot{1}\dot{5}$

15. $6.\dot{3}\dot{9}$

5. $0.\dot{2}\dot{4}$

16. $2.\dot{3}\dot{1}$

6. $0.\dot{1}\dot{7}$

17. $0.4\dot{3}$

7. $0.\dot{2}\dot{9}$

18. $0.2\dot{7}$

8. $0.0\dot{7}$

19. $0.5\dot{4}$

9. $0.0\dot{4}$

20. $0.32\dot{5}$

10. $0.\dot{3}\dot{5}$

21. $6.47\dot{6}$

11. $0.\dot{1}\dot{7}$

22. $2.12\dot{3}$

Compound interest

The compound interest formula is:

$$A = P\left(1 + \frac{r}{100}\right)^n$$

where A is the amount after n periods (eg months, years, half years etc), P is the principal (initial amount) and r is the interest rate per period.

Example: Find the amount which results when \$2000 is invested for 3 years at 5.3% per annum, where interest is compounded annually.

Solution:

$$\begin{aligned} A &= 2000 \left(1 + \frac{5.3}{100}\right)^3 \\ &= 2000 (1 + 0.053)^3 \\ &= \$2335.15 \end{aligned}$$

Note: To find the actual amount of interest, subtract the principal from this amount.

Determine the amount in an account if the interest is compounded when:

1. \$2000 is invested for 8 years at 3% per annum.
2. \$6000 is invested for 9 years at 4% per annum.
3. \$1000 is invested for 6 years at 3.5% per annum.
4. \$3000 is invested for 6 years at 6.36% per annum.
5. \$1000 is invested for 40 years at 3% per annum.
6. \$1000 is invested for 20 years at 7% per annum.
7. \$8000 is invested for 35 years at 7.5% per annum.

Find the amount of interest earned when:

8. \$200 is invested for 10 years at 3.75% per annum.
9. \$750 is invested for 12 years at 12.5% per annum.
10. \$7250 is invested for 16 years at 8.25% per annum.

Determine the amount of each investment when:

11. \$7000 is invested for 6 years at 10% per annum where interest is compounded each 6 months.
(Hint: the number of periods is 12 and the interest rate per period is 5%.)
12. \$6000 is invested for 7 years at 9% per annum where interest is compounded each 6 months.
13. \$1750 is invested for $3\frac{1}{2}$ years at 8.8% per annum where interest is compounded half yearly.
14. \$5000 is invested for 3 years at 6% per annum where interest is compounded each month.
15. \$7000 is invested for 42 months at 10% per annum where interest is compounded quarterly.

Superannuation

You may need the following formulae: $A = P(1 + \frac{r}{100})^n$ and $S_n = \frac{a(r^n - 1)}{r - 1}$

Example: Find the value of the superannuation fund of a woman who invested \$1000 at the beginning of each year for 10 years, at an interest rate of 6.3% per annum, if interest is paid at the end of each year and is compounded annually.

Solution: First \$1000 amounts to $1000(1.063)^{10}$

Second \$1000 amounts to $1000(1.063)^9$ etc

Last \$1000 amounts to $1000(1.063)^1$

Total = $1000(1.063^1 + 1.063^2 + \dots + 1.063^{10})$

= $1000 \times$ sum of a geometric series, where $a = 1.063$, $r = 1.063$ and $n = 10$

= $1000 \times \frac{1.063(1.063^{10} - 1)}{1.063 - 1}$

= 14 210.16

The value of her superannuation would be \$14 210.16.

- Determine the value of the superannuation fund of a man who invested \$2000 at the beginning of each year for 20 years, at an interest rate of 6.5% p.a. if interest is paid at the end of each year and compounded annually.
- David invests \$1500 at the beginning of each year in a superannuation fund. Determine the amount in the fund at the end of 25 years if interest is paid annually at the rate of 7.5% per annum.
- Freda invests \$1000 at the beginning of each year. She does this for each year for 30 years. Calculate the amount she will have if interest is compounded annually at the rate of 6.25% per annum.
- Lisa invests \$100 a month in a superannuation fund. Interest is paid monthly at the rate of 12% per annum. How much will she have at the end of 15 years.
- Freddie invests \$200 at the beginning of each month in a superannuation fund. Interest is paid monthly at the rate of 6% per annum. Determine the amount Freddie has in the scheme at the end of 15 years.
- A superannuation scheme offers two alternatives;
invest \$2400 at the beginning of each year for 20 years where interest is paid annually; or
invest \$200 at the beginning of each month for 20 years where interest is paid monthly.
If interest is paid at the rate of 9% per annum, determine which is the better alternative, and by how much.
- A superannuation fund offers two alternatives;
invest \$2500 at the beginning of each year for 20 years at 7.5% per annum; or
invest \$2500 at the beginning of each year for 10 years at 15% per annum.
If interest in both alternatives is paid annually, determine which is the better alternative, and by how much.

Geometric sequences and series review

You may need the following formulae:

$$T_n = ar^{n-1}, \quad S_n = \frac{a(r^n - 1)}{r - 1}, \quad S_n = \frac{a(1 - r^n)}{1 - r}, \quad S_\infty = \frac{a}{1 - r}$$

1. Is 1, -3, 9, ... a geometric series?
2. Is 6, 9, 12, ... a geometric sequence?
3. Determine the next two terms of the sequence 3, 9, 27, ...
4. Determine the next term of the sequence 5, 4, $3\frac{1}{5}$, ...
5. Determine the ninth term of the sequence 2, 6, 18, ...
6. The seventh term of geometric sequence is $7\frac{19}{32}$. If the first term is $\frac{1}{96}$, determine the common ratio.
7. Express 0.63 as a fraction in its simplest terms.
8. Determine the sum of the first fifteen terms of the series $\frac{1}{4} + \frac{1}{2} + 1 + \dots$
9. Determine the sum of the first nine terms of the series $25 + 5 + 1 + \dots$
10. Insert five terms between $\frac{2}{3}$ and 486 to form a geometric sequence.
11. Determine $12 + 6 + 3 + \dots$
12. How many terms of the series $\frac{2}{3} + \frac{1}{3} + \frac{1}{6} + \dots$ need to be added to obtain a total of $1\frac{5}{16}$?
13. The third term of a geometric sequence is 1 and the eighth term is $\frac{1}{32}$. Determine the first term and the common ratio.
14. The terms $2a$, $4b$, $8c$ form a geometric sequence. Write an expression for b , in terms of a and c (given a , b and c are positive).
15. Evaluate $\sum_{n=1}^9 8\left(\frac{1}{4}\right)^{n-1}$
16. The general term of a geometric sequence is given by $T_n = 3 \times 2^{n-2}$. Determine the first three terms and the common ratio.
17. An infinite series has a first term of 10 and a limiting sum of 15. Determine the common ratio.

Using the general expressions S_n and T_n in sequences and series

You may find these formulae useful: $T_n = a + (n-1)d$, $S_n = \frac{n}{2}(2a + (n-1)d)$,

$$T_n = ar^{n-1}, \quad S_n = \frac{a(r^n - 1)}{r - 1}, \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Examples: ① The n th term of a geometric sequence is given by $T_n = 3 \times 2^n$. Determine the first three terms and hence find the sum of the first seven terms.

Solution: $T_1 = 3 \times 2^1 = 6$ $T_2 = 3 \times 2^2 = 12$ $T_3 = 3 \times 2^3 = 24$

For the sequence, $a = 6$ and $r = 2$, $\therefore S_7 = \frac{6(2^7 - 1)}{2 - 1} = 762$

② Find the n th term of the sequence 4, 9, 13.

Solution: Sequence is arithmetic, with $a = 4$, $d = 5$.

$$\begin{aligned} \therefore T_n &= a + (n-1)d \\ &= 4 + (n-1)5 \\ &= 5n - 1 \end{aligned}$$

- The n th term of an arithmetic sequence is $T_n = 2 + 3n$. Find the first three terms.
- For an arithmetic sequence $T_n = 3 - 4n$. Determine the first three terms.
- A n th term of a geometric sequence is given by $T_n = 2^{2n-1}$. Determine the first three terms.
- The n th term of an arithmetic sequence is given by $6 + 2n$. Find the sum of the first fifteen terms.
- For a sequence $T_n = 4 + 2n$. Determine whether the sequence is arithmetic or geometric. Write down the 21st term.
- The general term of a geometric sequence is $T_n = 2^n \cdot 3^{n-1}$. Find the fifth and the seventh terms.
- For the geometric sequence $T_n = 3^{n-3}$, determine the first term and the common ratio.
- For the sequence $-3, 1, 5, \dots$, determine an expression for the T_n .
- Show that 148 is a term of the arithmetic sequence whose n th term is given by $T_n = 3n - 5$.
- Show that $\frac{25}{243}$ is a term of the geometric sequence $\frac{5^{n-3}}{3^n}$.
- The n th term of a sequence is given by $3^n + n$. Show that it is neither arithmetic nor geometric.
- The sum of an arithmetic series is given by $S_n = \frac{n}{2}(3n + 11)$. Find the sum of the first six terms.
- The sum of a geometric series is given by $S_n = 3(2)^{2-n} - 12$. Calculate the sum of the first seven terms.
- The n th term of an arithmetic sequence is $T_n = 4n - 11$. If there are 30 terms in the sequence, find the sum of the last ten terms.
- The sum of an arithmetic series is given by $S_n = \frac{n}{2}(5 + 3n)$. By finding S_1 and S_2 , determine the first term, the common difference and the seventh term.
- The sum of an arithmetic series is given by $S_n = 2n^2 - 24n$. How many terms must be added together to give a total of 128?
- The sum of a geometric series is $S_n = 128 - 2^{7-n}$. Find the first term and the common ratio.

Sequences and series review

You may need the following formulae: $T_n = a + (n-1)d$ $S_n = \frac{n}{2}(2a + (n-1)d)$ $S_n = \frac{n}{2}(a + l)$
 $T_n = ar^{n-1}$ $S_n = \frac{a(r^n - 1)}{r - 1}$ $S_n = \frac{a(1 - r^n)}{1 - r}$ $S_\infty = \frac{a}{1 - r}$

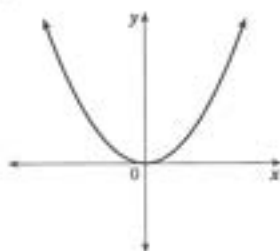
- Determine the twentieth term of the sequence 4, 8, 12, ...
- Determine the sum of the first sixteen terms of the series $6 + 10 + 14 + \dots$
- Calculate the sum of the first ten terms of the series $1 + 3 + 9 + \dots$
- Determine the sixteenth term of the sequence 8, 4, 2, ...
- What is the eighth term of the sequence 1, 5, 25, ...?
- Find the sum of the first ten terms of the series $1 + \frac{1}{2} + \frac{1}{4} + \dots$
- What is the tenth term of the sequence $1 + \frac{1}{2} + \frac{1}{4} + \dots$?
- Determine the sum of the following series: $4 + 2 + 1 + \dots$
- Write down the missing terms of the sequence ____, -11, -18, -25, ____.
- Write down the missing terms of the sequence ____, 10, ____, 18, 22.
- Determine the missing terms of the sequence ____, 9, 3, 1, ____.
- Determine the missing terms of the sequence ____, 18, -54, 162, ____.
- Write $0.\dot{7}\dot{1}$ as a fraction.
- Is 109 a term of 5, 13, 21, ...?
- What term of the sequence -2, -6, -18, ... is -118 098?
- How many terms of the series $-3 + 1 + 5 + \dots$ are needed to give a total of 228?
- Insert three numbers between 75 and 119 to form an arithmetic sequence.
- The general term of a sequence is $T_n = 2n + 1$. Determine the first three terms and the sum of the first fifteen terms.
- Evaluate $\sum_{n=1}^8 4 \times \left(\frac{1}{2}\right)^n$
- Evaluate $\sum_{n=1}^{\infty} 5 \times \left(\frac{1}{5}\right)^{n-1}$
- Which term of the sequence 6, 15, 24 ... is closest to 254?
- A geometric sequence has $T_8 = 49\ 152$ and $T_{11} = 3\ 145\ 728$. Determine the first three terms.
- Do the terms $\frac{1}{w-1}$, $\frac{1}{w}$, $\frac{1}{w+1}$ form an arithmetic sequence? Justify your answer.
- Find the sum of the first eighteen terms of the series $\sqrt{x} + \sqrt{4x} + \sqrt{9x} + \dots$

Sketching quadratics (basic parabolas)

Look at the following graphs and note how the sign and coefficient of x^2 affects the concavity. By letting $y = 0$, the x -intercept can be found, and the constant is the y -intercept.

Note: Don't forget to expand any brackets in an expression to obtain the y -intercept.

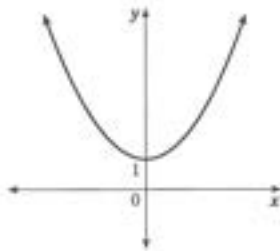
1 $y = x^2$



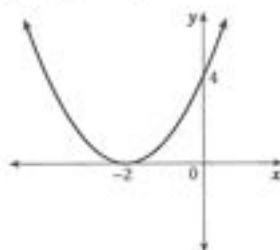
2 $y = -x^2$



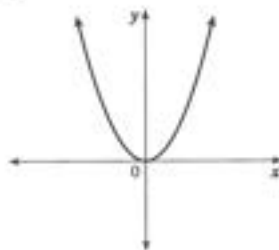
3 $y = x^2 + 1$



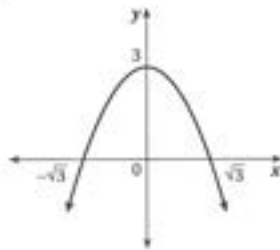
4 $y = (x + 2)^2$



5 $y = 3x^2$



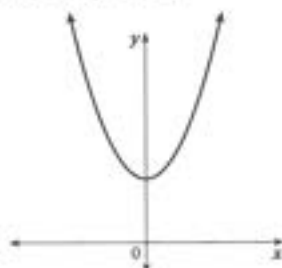
6 $y = -x^2 + 3$



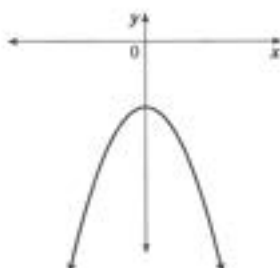
Note: To find the x -intercepts here, we need to solve the quadratic $-x^2 + 3 = 0$.

The graphs for each of the following equations have been drawn. Write down the missing x -intercept and y -intercept, where they exist.

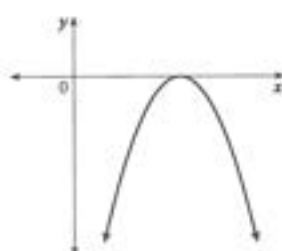
1. $y = x^2 + 4$



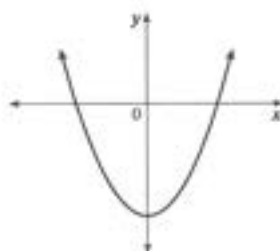
2. $y = -x^2 - 3$



3. $y = -(x - 4)^2$



4. $y = x^2 - 7$



Draw neat sketches of the following:

5. $y = 2x^2$

6. $y = -3x^2$

7. $y = x^2 + 3$

8. $y = x^2 - 2$

9. $y = -x^2 - 1$

10. $y = -x^2 + 1$

11. $y = (x + 3)^2$

12. $y = (x - 2)^2$

13. $y = -(x + 2)^2$

14. $y = -(x - 1)^2$

Match the following equations with the graphs below:

15. $y = -2x^2$

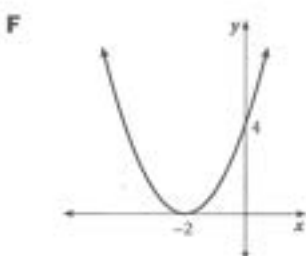
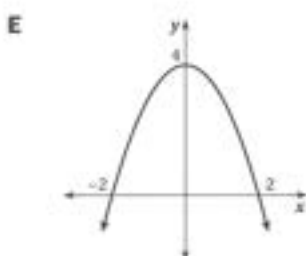
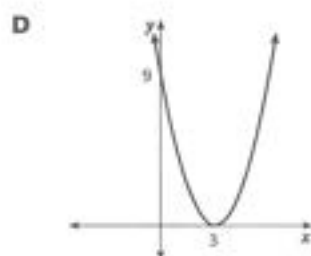
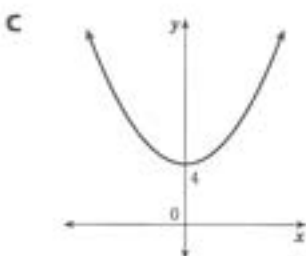
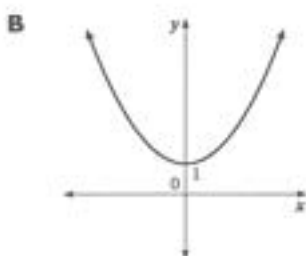
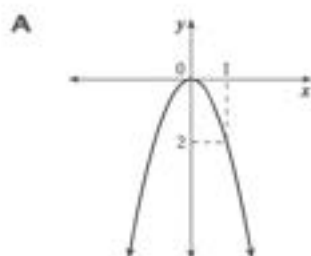
16. $y = (x - 3)^2$

17. $y = x^2 + 1$

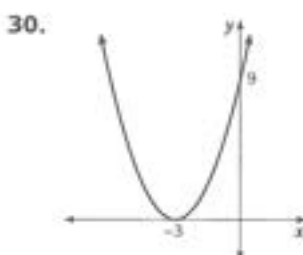
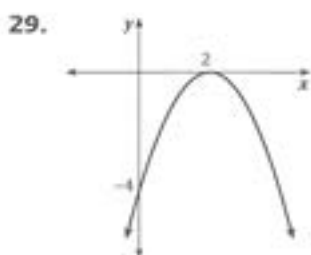
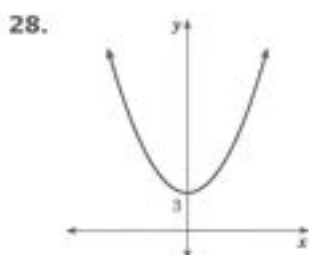
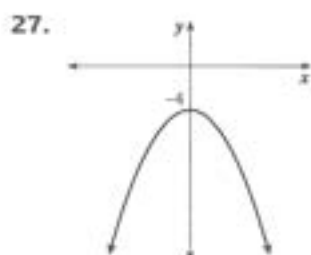
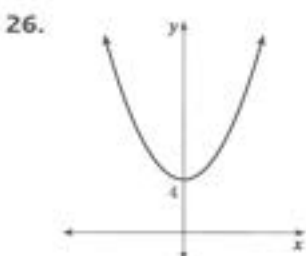
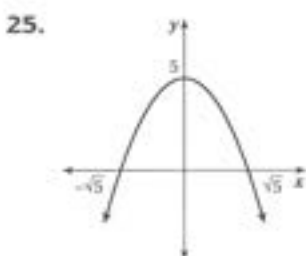
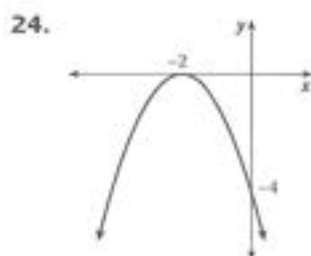
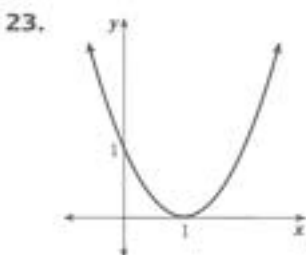
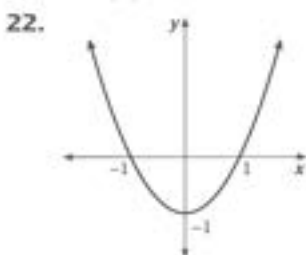
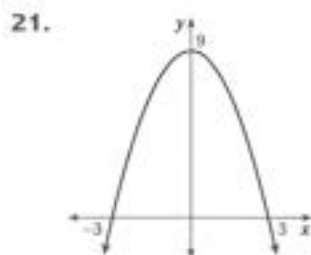
18. $y = (x + 2)^2$

19. $y = x^2 + 4$

20. $y = -x^2 + 4$



Write the equation for each of the following graphs:



Sketching parabolas using the intercepts and vertex

Example: Sketch the following quadratic function: $y = x^2 - 3x - 4$

Solution: Step 1: As the coefficient of x^2 is 1 (ie >0), the curve is concave up.

Step 2: The constant is -4 , therefore the y -intercept is -4 .

Step 3: To find the x -intercepts, let $y = 0$ and solve the quadratic equation.

$$x^2 - 3x - 4 = 0$$

$$(x - 4)(x + 1) = 0$$

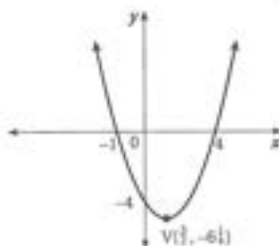
$$\therefore x = 4 \text{ or } x = -1$$

Note: Sometimes it is difficult, or impossible, to factorise the equation and it may not be essential to the question.

Step 4: For the general equation $y = ax^2 + bx + c$, the equation of the axis of symmetry is $x = \frac{-b}{2a}$
 $\therefore x = \frac{3}{2} \text{ or } x = 1\frac{1}{2}$

Note: This is the x -coordinate of the vertex or turning point of the parabola.

Step 5: The y -coordinate of the vertex is $y = \left(\frac{3}{2}\right)^2 - 3 \times \frac{3}{2} - 4$
 $\therefore y = -6\frac{1}{4}$



State the x -intercepts, the y -intercepts and the coordinates of the vertex for the following:

1. $y = x^2 + 4x + 3$

2. $y = x^2 - 6x + 5$

3. $y = x^2 - 2x$

4. $y = -x^2 + 3x$

5. $y = x^2 + 5x + 6$

6. $y = x^2 + 4x - 12$

7. $y = -x^2 - 2x + 3$

8. $y = -x^2 - 7x - 10$

Draw neat sketches of the following quadratic functions, showing x -intercepts and y -intercept

9. $y = x^2 + x - 12$

10. $y = x^2 - 3x - 10$

11. $y = x^2 + 5x - 6$

12. $y = 2x^2 - 7x + 3$

13. $y = 3x^2 + 17x + 10$

14. $y = 2x^2 + 7x + 6$

15. $y = 4x^2 - 4x - 3$

16. $y = 2x^2 - 13x + 15$

17. $y = -x^2 - 2x + 8$

18. $y = -x^2 + 2x + 3$

19. $y = -x^2 + x + 12$

20. $y = -2x^2 + 5x - 2$

Solving quadratic equations using the formula

When it is difficult or impossible to factorise quadratic equations, use the quadratic formula:

$$\text{for the general quadratic equation, } ax^2 + bx + c = 0, \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

When using this formula, the answer could be (i) rational, (ii) irrational (a surd), (iii) a decimal approximation or (iv) impossible to solve.

Example: Solve the equation $x^2 + 6x + 2 = 0$, leaving the answer in simplest surd form.

$$a = 1, b = 6, c = 2$$

$$\begin{aligned} x &= \frac{-6 \pm \sqrt{6^2 - 4 \times 1 \times 2}}{2 \times 1} \\ &= \frac{-6 \pm \sqrt{28}}{2} \\ &= \frac{-6 \pm 2\sqrt{7}}{2} \\ &= -3 \pm \sqrt{7} \end{aligned}$$

Solve the following quadratic equations, using the quadratic formula. Where necessary, leave answers in simplest surd form.

1. $x^2 + 3x + 1 = 0$

2. $x^2 - 3x - 5 = 0$

3. $2x^2 + 7x - 3 = 0$

4. $-x^2 + 4x + 1 = 0$

5. $x^2 + 4x - 6 = 0$

6. $x^2 - 6x - 5 = 0$

7. $2x^2 + 5x + 2 = 0$

8. $6x^2 - 7x + 2 = 0$

9. $2x^2 + 5x - 1 = 0$

10. $-2x^2 + 3x + 4 = 0$

Solve the following, giving your answer correct to two decimal places:

11. $x^2 + x - 3 = 0$

12. $2x^2 + 6x + 1 = 0$

13. $2 + x^2 + 4x = 0$

14. $6 - 3x - 2x^2 = 0$

15. $-4x^2 + 2 + x = 0$

16. $2x^2 - 2 - 5x = 0$

17. $3x^2 - 8x + 5 = 0$

18. $-x^2 - 7x - 4 = 0$

19. $-2x^2 + 6x + 3 = 0$

20. $-1 - 5x^2 + 12x = 0$

The discriminant of a quadratic equation

The formula for solving the quadratic equation $ax^2 + bx + c = 0$ is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The term under the square root sign, ie. $b^2 - 4ac$, is called the **discriminant** and is denoted by the symbol Δ .

Hence, $\Delta = b^2 - 4ac$

The value of Δ determines the number and type of **roots** (solutions) of the quadratic equation.

If $\Delta = 0$ there are two equal and rational roots.

If $\Delta > 0$ there are two unequal roots.

If $\Delta < 0$ there are no real roots (unreal roots).

Also,

If Δ is a perfect square the roots are rational.

If Δ is not a perfect square the roots are irrational (ie a surd).

Determine the nature of the roots of the following quadratic equations (ie whether they are real or unreal, equal or unequal, rational or irrational):

1. $x^2 + 4x + 4 = 0$

7. $4x^2 + 16x + 16 = 0$

2. $x^2 - 3x + 4 = 0$

8. $3x^2 + 4x + 2 = 0$

3. $2x^2 - 3x - 1 = 0$

9. $12x^2 - 26x + 10 = 0$

4. $2x^2 - 2x - 12 = 0$

10. $x^2 - 4x + 9 = 0$

5. $x^2 - 6x + 9 = 0$

11. $9x^2 - 30x + 25 = 0$

6. $2x^2 + 4x + 12 = 0$

12. $3x^2 - 13x + 4 = 0$

13. If $2x^2 - 5x + k = 0$ has equal roots, determine the value of k .

14. The quadratic equation $kx^2 + 3x + 2 = 0$ has unreal roots. Determine the values of k .

15. Determine the value of k so that the equation $x^2 - 7x + k = 0$ has two unequal roots.

16. If the equation $kx^2 + 3kx + 9 = 0$ has two equal roots, determine the value of k .

17. For what values of k does the graph of the equation $x^2 - 6x + 3(k + 1) = y$ intersect the x -axis at two points?

18. Show that the equation $20x^2 + 3x - 9 = 0$ has rational roots.

Sums and products of roots of quadratic equations

For the quadratic equation $ax^2 + bx + c = 0$, whose roots are α and β ,

the sum of the roots is found using: $\alpha + \beta = \frac{-b}{a}$

and the product of the roots is found using: $\alpha\beta = \frac{c}{a}$

$$\text{Note: } \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

Given the roots α and β , the quadratic equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

Example: If the sum of the roots of a quadratic equation is $\frac{2}{3}$ and the product of its roots is $\frac{1}{4}$, find the equation with integer coefficients.

$$x^2 - \frac{2}{3}x + \frac{1}{4} = 0$$

$$\therefore 12x^2 - 8x + 3 = 0 \text{ is the equation}$$

If the roots of the quadratic equation $2x^2 - 4x + 3 = 0$ are α and β , find the value of:

- | | |
|---------------------|-------------------------|
| 1. $\alpha + \beta$ | 3. $(\alpha + \beta)^2$ |
| 2. $\alpha\beta$ | 4. $\alpha^2 + \beta^2$ |

If the roots of the equation $x^2 - 8x - 4 = 0$ are α and β , determine:

- | | |
|---|-------------------------|
| 5. $3\alpha + 3\beta$ | 7. $\alpha^2 + \beta^2$ |
| 6. $\frac{1}{\alpha} + \frac{1}{\beta}$ | 8. $(\alpha - \beta)^2$ |

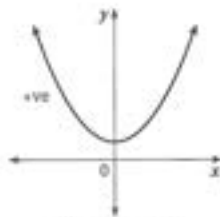
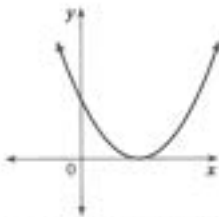
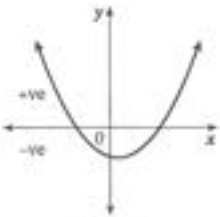
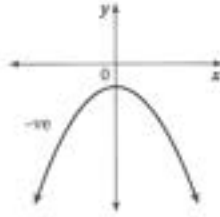
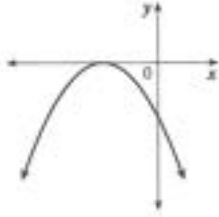
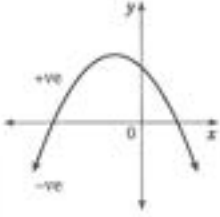
If α and β are the roots of the equation $2x^2 + 3x - 3 = 0$, find:

- $(\alpha + 4)(\beta + 4)$
- $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$
- Determine the simplest quadratic equation (with integer coefficients) where the sum of the roots is 4 and the product of the roots is 3.
- Determine the simplest quadratic equation (with integer coefficients) where the product of the roots is 8 and the sum of the roots is -4 .
- Find a quadratic equation (with integer coefficients) where the sum of its roots is $\frac{1}{5}$ and the product of its roots is -5 .
- Find a quadratic equation (with integer coefficients) where the product of its roots is $\frac{1}{4}$ and the sum of its roots is $\frac{1}{3}$.
- Find a quadratic equation (with integer coefficients) where the product of its roots is $-\frac{1}{2}$ and the sum of its roots is $-\frac{7}{6}$.
- Find a quadratic equation with roots 6 and -2 .

The discriminant and the parabola

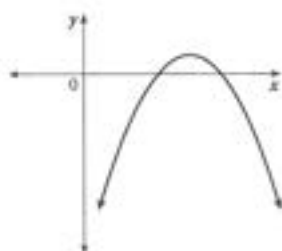
For the quadratic function $y = ax^2 + bx + c$, the discriminant, $\Delta = b^2 - 4ac$, and the coefficient, a , determine the sign of the function.

- When the function is:
- i positive, the parabola is above the x -axis
 - ii zero, the parabola is on the x -axis
 - iii negative, the parabola is below the x -axis

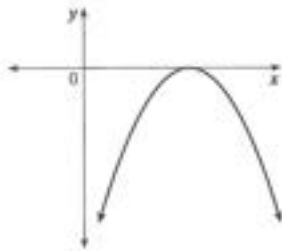
	$\Delta < 0$	$\Delta = 0$	$\Delta > 0$
$a > 0$	 Positive definite	 +ve except for 0 at the vertex	 Indefinite
$a < 0$	 Negative definite	 -ve except for 0 at the vertex	 Indefinite

For each graph, state the sign of the discriminant and the coefficient of x^2 .

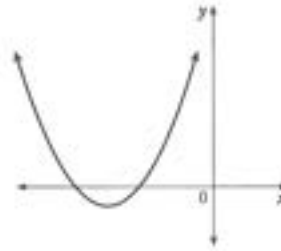
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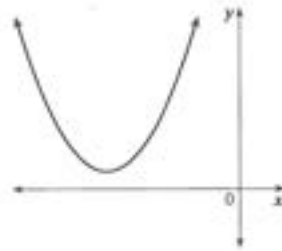
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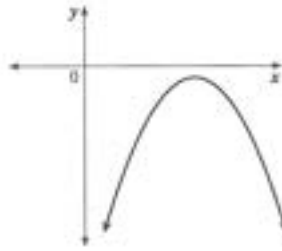
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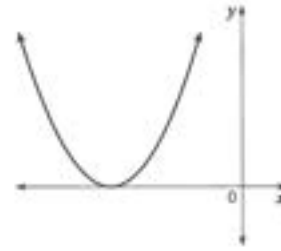
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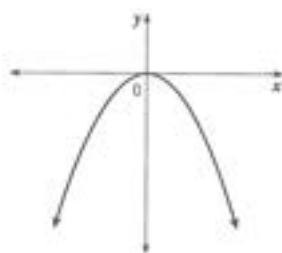
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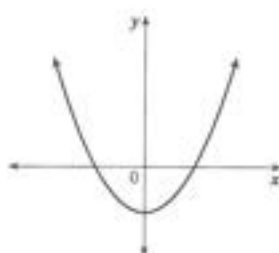
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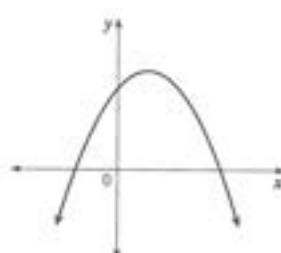
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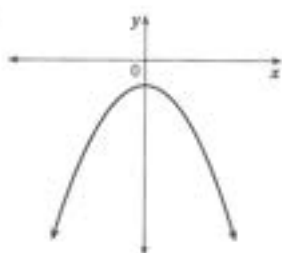
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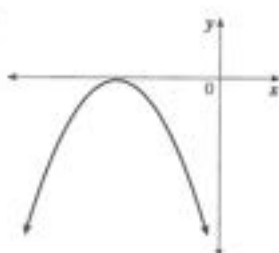
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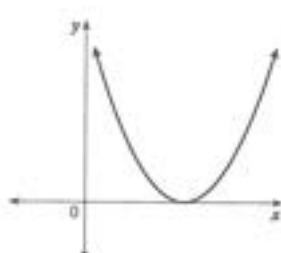
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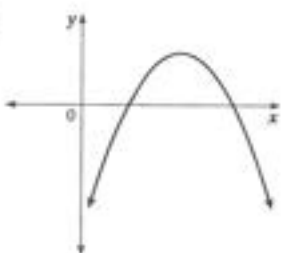
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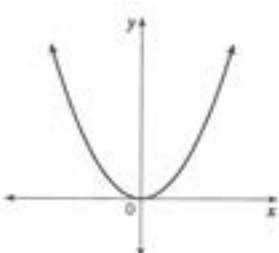
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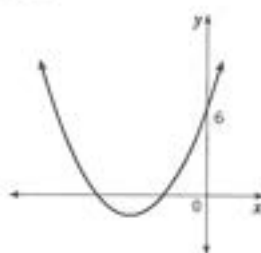


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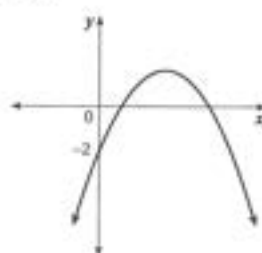


Which of the two equations is represented by the graph:

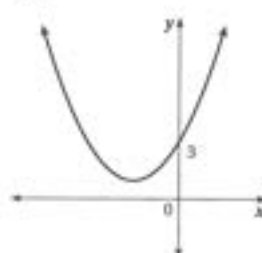
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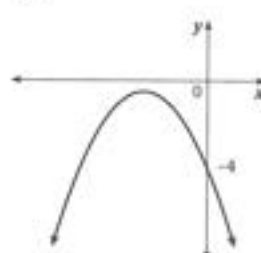
16.



17.



18.



A) $y = x^2 + 4x + 6$

B) $y = x^2 + 9x + 6$

A) $y = -2x^2 + 5x - 2$

B) $y = -2x^2 + 3x - 2$

A) $y = x^2 + 3x + 3$

B) $y = x^2 + 4x + 3$

A) $y = -x^2 - x - 4$

B) $y = -x^2 - 5x - 4$

19. Show that $x^2 + x + 3$ is always positive (positive definite).

20. Show that $2x^2 + 4x + 2$ is zero for one value of x and positive for all other values.

21. Show that $x^2 + 4x + 3$ is indefinite.

22. For what values of k is $2x^2 + x + k$ positive definite?

23. For what values of m is $mx^2 + 3x + 4$ indefinite?

24. For what values of a is $ax^2 + 2x + 3$ positive definite?

25. For what values of k is $kx^2 + 4x + k$ positive definite?

26. Find the positive value of p for which $x^2 - (p + 1)x + (p + 1) = 0$ has two equal roots.

Roots of quadratic equations review

Summary: Some of the following formulae may be useful:

$$\Delta = b^2 - 4ac, \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad \alpha + \beta = \frac{-b}{a}, \quad \alpha\beta = \frac{c}{a}, \quad x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

where α and β are the roots of the equation $ax^2 + bx + c = 0$.

1. Show that $9x^2 + 12x + 4 = 0$ has two equal roots.
2. Without solving the equation $x^2 + x + 2 = 0$, show that it has no real roots.
3. Show that $x^2 + 2x - 3 = 0$ has two unequal roots.
4. Show that $-x^2 + 3x - 5 = 0$ does not cut the x -axis.

The roots of $x^2 - 5x + 3 = 0$ are α and β . Find:

5. $\alpha + \beta$
6. $\alpha\beta$
7. $\alpha^2 + \beta^2$
8. $(\alpha - \beta)^2$
9. $(\alpha - 3)(\beta - 3)$
10. $\frac{1}{\alpha} + \frac{1}{\beta}$
11. Write down the simplest quadratic equation in which the sum of the roots is 6 and the product is -4.
12. Write down the simplest quadratic equation (with integer coefficients) where the sum of the roots is $-\frac{3}{2}$ and the product of the roots is $\frac{2}{3}$.
13. The roots of a quadratic equation add to 12. If one root is twice the other, write down a possible equation.
14. If the roots of $x^2 + 3x + k = 0$ are equal, determine the value of k .
15. One root of a quadratic equation is 3 bigger than the other. If the equation is $4x^2 - 16x + k = 0$, determine the value of k .

The roots of the equation $x^2 - 4x - 3 = 0$ are α and β . Find:

16. $\alpha + \beta$
17. $(\alpha\beta)^2$
18. $(\alpha + 5)(\beta + 5)$
19. $\alpha^2 + \beta^2 + \alpha\beta$
20. $\frac{1}{3\alpha} + \frac{1}{3\beta}$
21. $\frac{1}{\alpha + 1} + \frac{1}{\beta + 1}$

Solving quadratic equations by completing the square

Example: Solve the equation $x^2 + 6x + 1 = 0$ by 'completing of square' method

$$x^2 + 6x + 1 = 0$$

$$x^2 + 6x = -1$$

$$x^2 + 6x + (3^2) = -1 + (3^2) \quad (\text{halve the coefficient of } x, \text{ square it and add to both sides})$$

$$(x + 3)^2 = -1 + 9$$

$$(x + 3) = \pm \sqrt{8}$$

$$\therefore x = -3 \pm 2\sqrt{2}$$

Note: If the coefficient of x is not 1, then divide throughout by the coefficient before completing the square.

Write down the missing numbers in the following equations:

1. $x^2 + \square x + 4 = (x + 2)^2$

3. $x^2 - 18x + \square = (x - \triangle)^2$

2. $x^2 + \square x + 9 = (x + \triangle)^2$

4. $4x^2 + \square x + \triangle = (2x + 6)^2$

5. Complete the following example by filling in the missing numbers:

$$2x^2 + 6x - 1 = 0$$

$$x^2 + \square x = \frac{1}{2}$$

$$x^2 + 3x + \left(\frac{3}{2}\right)^2 = \frac{1}{2} + \square$$

$$(\square + \triangle)^2 = \frac{11}{4}$$

$$x + \frac{3}{2} = \square$$

$$\therefore x = \frac{-3 \pm \sqrt{11}}{2}$$

Solve the following equations, by completing the square:

6. $x^2 + 4x - 1 = 0$

11. $x^2 + 4x - 7 = 0$

7. $x^2 + 6x + 2 = 0$

12. $2x^2 + 8x + 7 = 0$

8. $x^2 - 3x + 1 = 0$

13. $2x^2 + 9x + 2 = 0$

9. $x^2 - 5x + 2 = 0$

14. $3x^2 + 3x - 4 = 0$

10. $x^2 + 7x - 4 = 0$

15. $5x^2 - 10x - 6 = 0$

Quadratic equations and functions review

To solve a quadratic equation, try to factorise first. If this is not possible or looks too difficult, use either the 'completing the square' method or the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example: Solve by 'completing the square' method, the quadratic equation $x^2 + 7x - 4 = 0$.

Solution: $x^2 + 7x + \left(\frac{7}{2}\right)^2 = 4 + \left(\frac{7}{2}\right)^2$

$$\left(x + \frac{7}{2}\right)^2 = \frac{16 + 49}{4}$$

$$\left(x + \frac{7}{2}\right) = \pm \sqrt{\frac{65}{4}}$$

$$\therefore x = -\frac{7}{2} \pm \frac{\sqrt{65}}{2}$$

$$\therefore x = \frac{-7 \pm \sqrt{65}}{2}$$

Solve the following quadratic equations. Where necessary, write the answer in simplest surd form.

1. $x^2 - 25 = 0$

2. $x^2 - 6 = 0$

3. $x^2 + 4x = 0$

4. $2x^2 + 6x = 0$

5. $x^2 - x + 12 = 0$

6. $x^2 - x - 6 = 0$

7. $x^2 + 8x + 16 = 0$

8. $x^2 - 10x + 25 = 0$

9. $2x^2 + 11x + 12 = 0$

10. $6x^2 + 17x + 12 = 0$

11. $x^2 + x - 3 = 0$

12. $x^2 + 7x + 5 = 0$

13. $2x^2 + 3x - 5 = 0$

14. $2x^2 - 9x + 3 = 0$

Draw neat sketches of the following:

15. $y = x^2 + 5x + 6$

16. $y = (x + 3)^2$

17. $y = 2x^2 - 5x - 3$

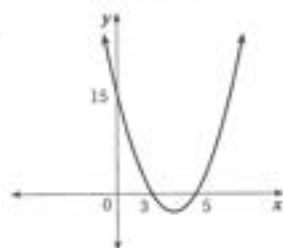
18. $y = 1 - x^2$

19. Determine the x -intercepts and y -intercept for the function $y = 2x^2 + 5x + 2$.

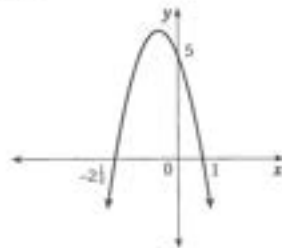
20. Determine the coordinates of the vertex of the parabola with equation $y = x^2 + 6x + 1$.

Write down the equations of the parabolas sketched below:

21.



22.

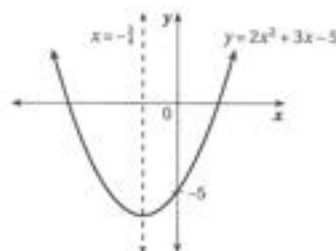


Graphing quadratics using the discriminant

In order to sketch a graph of a quadratic, it is not always necessary to determine the x -intercepts. Generally the nature of the question will indicate whether the x -intercepts need to be found.

Example: Sketch the graph of $y = 2x^2 + 3x - 5$.

- i) $\Delta = b^2 - 4ac$
 $= 9 - 4 \times 2 \times (-5)$
 $= 49$
 $\therefore$ cuts the x -axis in 2 places
- ii) $a = 2$
 $\therefore$ curve is concave up
- iii) Equation of axis is $x = -\frac{b}{2a}$
 $= -\frac{3}{4}$

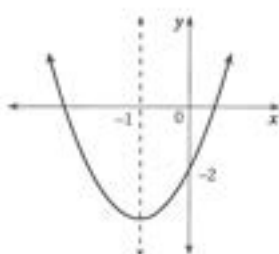


Sketch the following equations:

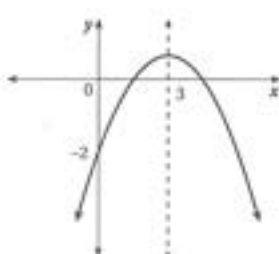
1. $y = x^2 + x + 2$
2. $y = -x^2 + 6x - 2$
3. $y = -x^2 - x + 1$
4. $y = x^2 - 4x + 2$
5. $y = -x^2 + 2x - 5$
6. $y = -x^2 - 7x + 12$
7. $y = 2x^2 - 3x - 4$
8. $y = 16 + 4x + x^2$
9. $y = 3x^2 + 2x - 3$
10. $y = -4x^2 + x + 1$

Match the following sketches with the correct equation:

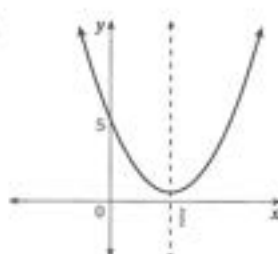
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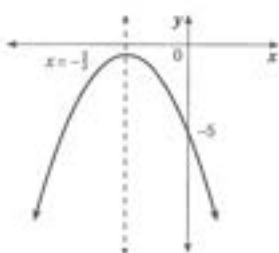
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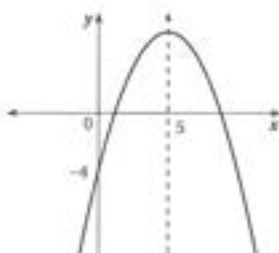
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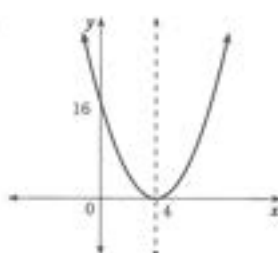
14.



15.



16.



A $y = x^2 + 2x - 2$

B $y = -2 + 3 - \frac{1}{2}x^2$

C $y = x^2 - 3x + 5$

D $y = -2x^2 - 6x + 5$

E $y = -x^2 - 3x - 5$

F $y = -x^2 - 5x - 4$

G $y = -x^2 + 10x - 4$

H $y = x^2 + 8x + 16$

I $y = x^2 - 8x + 16$

The parabola as a locus: vertex the origin

The definition of a parabola, using the concept of a locus is:

The path traced out by a point as it moves, so that its distance from a fixed point (the focus) is equal to its perpendicular distance from a fixed line (the directrix).

The equation of a parabola with its vertex (turning point) at the Origin is:

$$x^2 = 4ay \text{ (concave up) or } x^2 = -4ay \text{ (concave down)}$$

where the axis is the y -axis and a is the focal length (distance from the vertex to the focus).

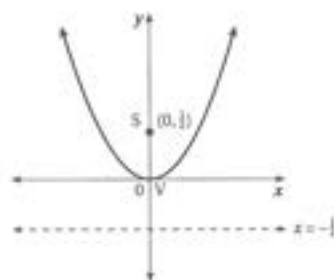
Example: For $x^2 = 2y$

i) the vertex, V , is $(0, 0)$

ii) to find the focus let $4a = 2$

$\therefore a = \frac{1}{2}$ ie the focus, S , has coordinates $(0, \frac{1}{2})$

iii) the directrix is $x = -\frac{1}{2}$



Write down the focus and the equation of the directrix for the following:

1. $x^2 = 4y$

4. $x^2 = 12y$

7. $x^2 = -3y$

10. $x^2 = -5y$

2. $x^2 = 8y$

5. $x^2 = y$

8. $x^2 = -2y$

3. $x^2 = 16y$

6. $x^2 = -8y$

9. $x^2 = -4y$

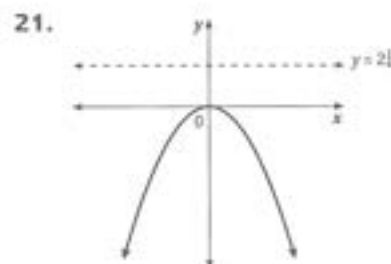
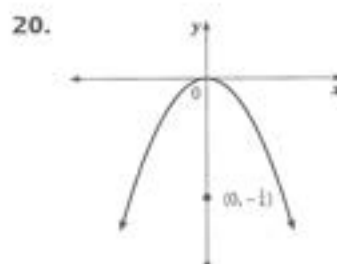
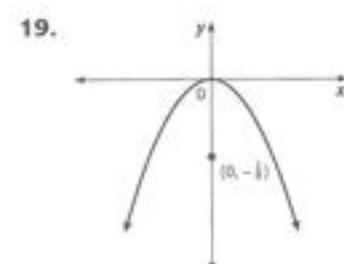
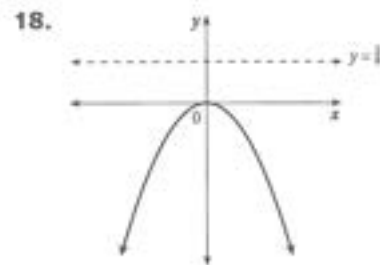
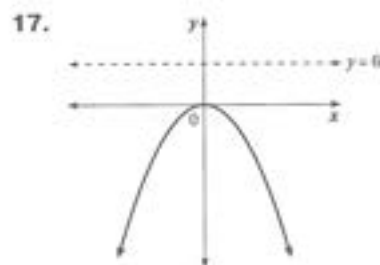
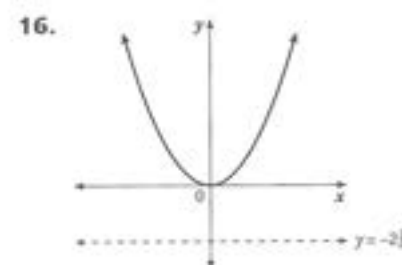
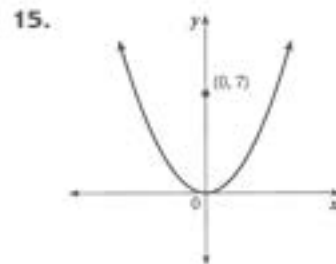
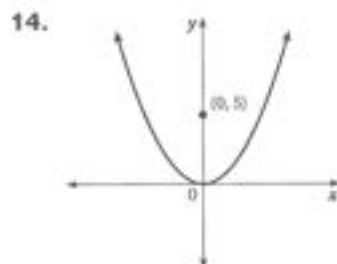
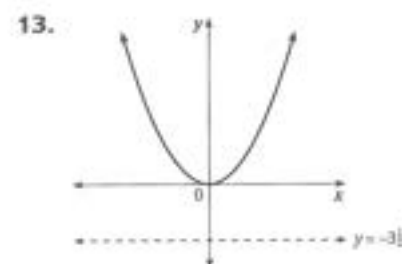
Draw neat sketches of the following, indicating the focus and the directrix.

11. $x^2 = -12y$

12. $\frac{1}{4}x^2 = -y$

From the following diagrams, write down the equation of the parabola.

Note: either the coordinates of the focus or the equation of the directrix has been shown.



The parabola as a locus: vertex not the origin (1)

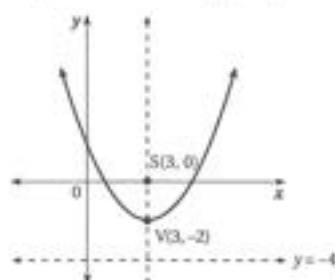
The definition of a parabola, using the concept of a locus is:

The path traced out by a point as it moves, so that its distance from a fixed point (the focus) is equal to its perpendicular distance from a fixed line, (the directrix).

The equation of a parabola whose axis is parallel to the y-axis, with vertex (h, k) and a focal length of a is: $(x - h)^2 = 4a(y - k)$ concave upwards

$(x - h)^2 = -4a(y - k)$ concave downwards

Example: For $(x - 3)^2 = 8(y + 2)$ the vertex is $(3, -2)$, the focal length is 2, the focus is $(3, 0)$ and the equation of the directrix is $y = -4$.

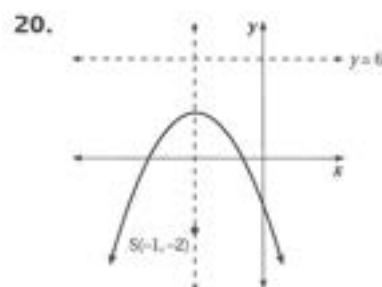
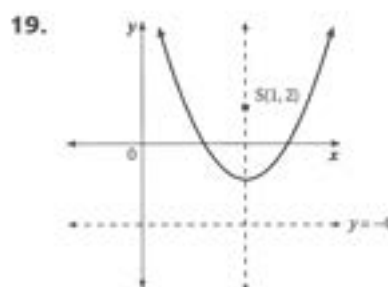
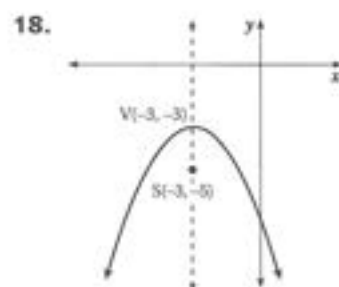
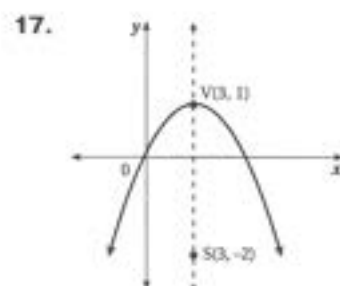
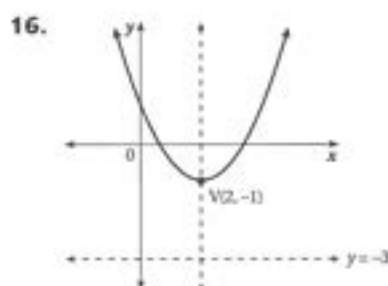
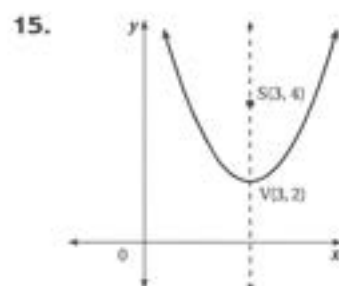


It often helps to sketch the parabola to establish the focus and the directrix.

Draw neat sketches of the following parabolas. Write down the coordinates of the vertex (V), the focus (S) and the equation of the directrix.

1. $(x - 3)^2 = 4(y - 1)$
2. $(x - 2)^2 = 12(y - 3)$
3. $(x - 5)^2 = 16(y + 3)$
4. $(x + 3)^2 = 4(y - 2)$
5. $(x + 2)^2 = -8(y + 3)$
6. $(x + 4)^2 = -12(y + 1)$
7. $(x - 1)^2 = 2(y + 3)$
8. $(x + 1)^2 = -6(y - 1)$
9. $x^2 = 8(y + 4)$
10. $-(y + 1) = (x - 3)^2$
11. $(x - 2)^2 = 2(y - \frac{1}{2})$
12. $(x - 2)^2 = -2y$
13. $(x - 3)^2 = 4y - 16$
14. $(x + 1)^2 = 4 - 8y$

Write down the equations of the following parabolas:



The parabola as a locus: vertex not the origin (2)

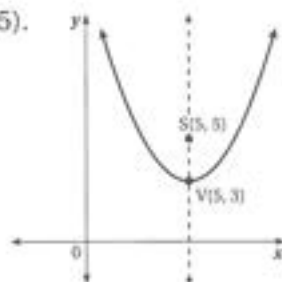
The equation of a parabola, given the coordinates of the vertex and the coordinates of the focus or the equation of the directrix, is: $(x - h)^2 = \pm 4a(y - k)$, where (h, k) is the vertex and the focal length is a . (positive—concave up; negative—concave down)

Example: Find the equation of the parabola with vertex $(5, 3)$ and focus $(5, 5)$.

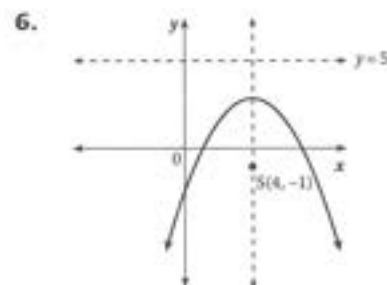
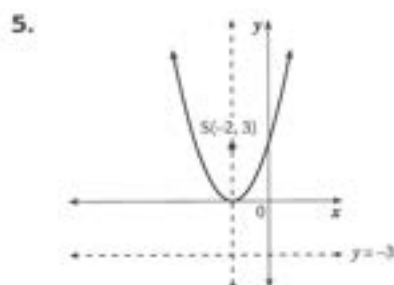
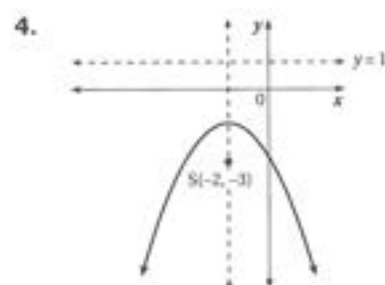
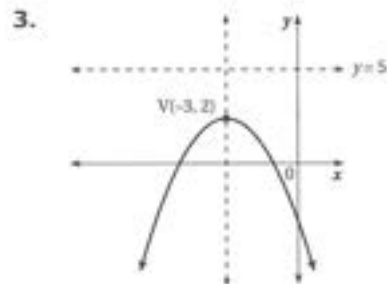
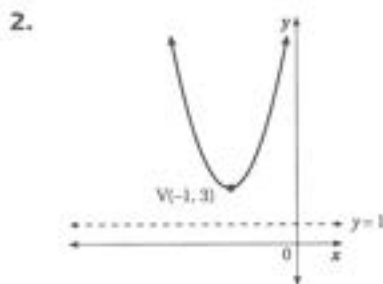
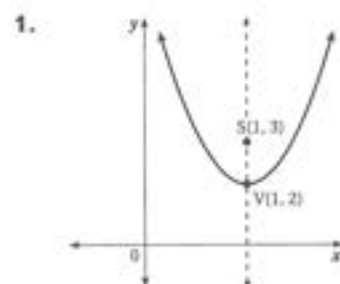
Solution: Plot these two points, then decide where to draw the parabola, remembering that the focus is always 'inside' the parabola.

From the sketch, the focal length is 2, and the equation is:

$$(x - 5)^2 = 4 \times 2(y - 3) \text{ or } (x - 5)^2 = 8(y - 3)$$



Write down the equations of the following parabolas:



- What is the equation of the parabola with vertex $(-2, -3)$ and focus $(-2, 1)$?
- A parabola has a focus at $(3, 6)$ and a vertex at $(3, 4)$. Write down its equation.
- A parabola has a focus at $(2, -2)$ and the equation of its directrix is $y = 4$. Write down the equation of the parabola.
- Find the equation of the parabola if the equation of the directrix is $y = -3$ and focus is $(2, 3)$.
- The equation of the directrix of a parabola is $y = -2$, and its vertex is $(3, 4)$. Determine the equation of the parabola.
- Determine the equation of the parabola with focus $(-2, 1)$ and directrix $y = -2$.
- Find the equation of the parabola with a focal length of 3 units, vertex $(2, -1)$ and concave down.
- Write down the equation of the parabola whose focus is $(4, 5)$ and whose directrix is 4 units below the focus.
- Find the equation of the parabola(s) whose vertex is $(3, -1)$ and whose directrix is 5 units from the vertex.
- Write down the equation of the parabola where the focus is $(3, 1)$ and the directrix is 6 units above the focus.

The circle as a locus

The circle represents the locus of a point that moves so that it is always equidistant (radius) from a fixed point (centre).

The equation of a circle, centre the origin, radius r is: $x^2 + y^2 = r^2$

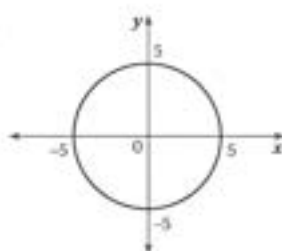
The equation of a circle, centre (h, k) , radius r is: $(x - h)^2 + (y - k)^2 = r^2$

Write down the coordinates of the centre and the length of the radius for the following:

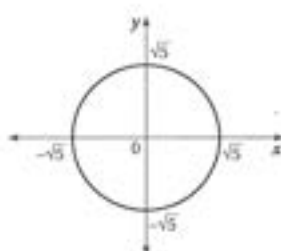
1. $x^2 + y^2 = 4$
2. $x^2 + y^2 = 9$
3. $x^2 + y^2 = 1$
4. $x^2 + y^2 = 3$
5. $(x - 3)^2 + (y - 4)^2 = 16$
6. $(x + 1)^2 + (y + 2)^2 = 4$
7. $(x + 2)^2 + (y + 3)^2 = 25$
8. $(x - 2)^2 + (y + 3)^2 = 8$
9. $(x - 2)^2 + (y + 1)^2 = \frac{1}{4}$
10. $(x + 3)^2 + (y - 2)^2 = \frac{4}{9}$

Write down the equation of the following:

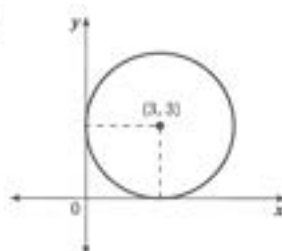
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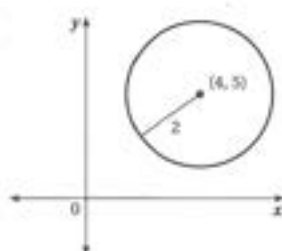
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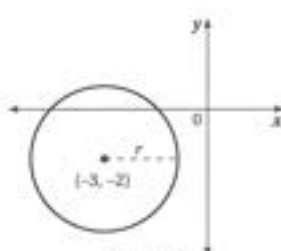
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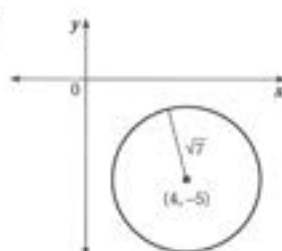
14.



15.



16.



17. Determine the equation of a circle with centre $(0, 0)$ and with the point $(3, 4)$ lying on the circumference.
18. Determine the equation of the circle, centre $(0, 0)$, given that the point $(2, 3)$ lies on the circumference.
19. The point $(6, 7)$ lies on the circumference of a circle with centre $(3, 4)$. Determine the equation of the circle.
20. Is the point $(0, 3)$ on the circumference of a circle, centre $(2, 3)$ and radius 2?
21. Is the point $(-5, 8)$ inside or outside the circle, centre $(-1, 4)$, with a radius of 5?

Finding the focus, vertex and directrix by completing the square

The equation of a parabola, vertex (h, k) and focal length a , concave up is: $(x - h)^2 = 4a(y - k)$
 concave down is: $(x - h)^2 = -4a(y - k)$

Sometimes the equation of the parabola is given in an expanded form, eg $x^2 - 6x + 1 = 4y$.

Example: Given the equation of the parabola $x^2 - 6x + 1 = 4y$, find the coordinates of the vertex and the focus and the equation of the directrix.

Solution: $x^2 - 6x + 1 = 4y$

Complete the square (ie halve the coefficient of x , square it, then add it to both sides)

$$(x^2 - 6x + 9) + 1 = 4y + 9$$

$$(x - 3)^2 = 4y + 8$$

$$(x - 3)^2 = 4(y + 2)$$

$\therefore$ the vertex is $(3, -2)$

The focal length is 1, so the focus is $(3, -1)$ and the equation of the directrix is $y = -3$.

Note: A neat sketch of the parabola is also helpful in establishing the above.

Determine the coordinates of the vertex and focus and the equation of the directrix for the following:

1. $x^2 - 4x + 28 = 8y$

2. $x^2 - 6x = 4y - 5$

3. $x^2 - 16x + 4y + 16 = 0$

4. $x^2 - 6x - 2y = 3$

5. $x^2 + 4x = 8y$

6. $x^2 - 14x + 9 + 4y = 0$

7. $x^2 - 2x - 1 = 4y$

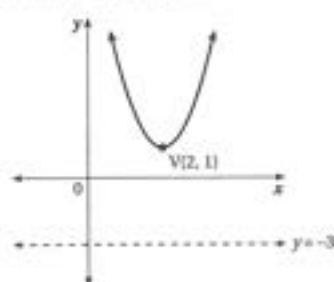
8. $x^2 + 6x + 8y + 5 = 0$

9. $x^2 + 6y + 10x + 31 = 0$

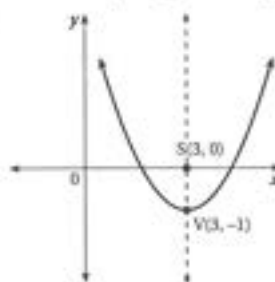
10. $x^2 + 8x - y + 14 = 0$

Determine the equation of each of the following parabolas and, by expanding, make y the subject.

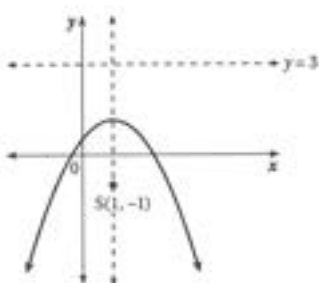
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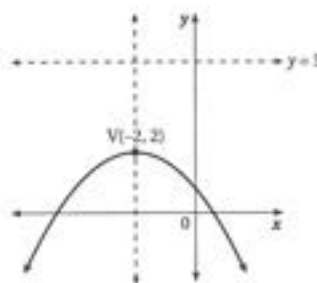
12.



13.



14.



Equations of circles in expanded form

The general form for the equation of a circle is: $x^2 + y^2 + 2hx + 2ky + c = 0$

Example: Show that $x^2 + y^2 + 6x + 4y + 9 = 0$ is a circle, and write down the coordinates of the centre and the length of the radius.

Solution: $x^2 + y^2 + 6x + 4y + 9 = 0$

$$x^2 + 6x + y^2 + 4y = -9$$

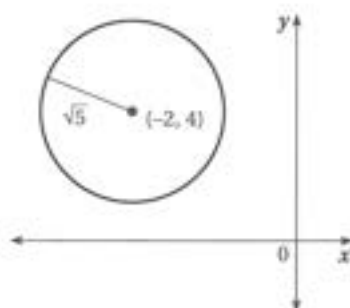
Complete the square (ie halve the coefficients of x and y , square them, then add them to both sides)

$$x^2 + 6x + (3)^2 + y^2 + 4y + (2)^2 = -9 + 9 + 4$$

$$\therefore (x + 3)^2 + (y + 2)^2 = 4 \quad (\text{ie in the form } (x - h)^2 + (y - k)^2 = r^2)$$

This is the equation of a circle, centre $(-3, -2)$, radius 2 units.

1. A circle has a centre $(3, 4)$ and a radius of 4 units. Write down the equation of the circle, and then, by expanding and simplifying, find the equation in general form.
2. Write down the equation of a circle in general form, if the centre is $(-5, 2)$ and the radius is 1 unit.
3. Write down the equation of a circle in general form, if it has a centre $(0, -3)$ and a $r = \sqrt{3}$.
4. Determine the equation of the circle below, in general form.



Determine the radius and centre of each of the following circles:

5. $x^2 - 6x + y^2 - 4y - 23 = 0$

10. $x^2 + y^2 + 6y = 0$

6. $x^2 - 4x + y^2 + 2y + 4 = 0$

11. $x^2 + 6x + y^2 - 2y - 15 = 0$

7. $x^2 + 10x + y^2 - 8y + 16 = 0$

12. $x^2 - 2x + y^2 - 4y + 4 = 0$

8. $x^2 + 2x + y^2 + 8y + 1 = 0$

13. $x^2 - x + y^2 - 2y - 14\frac{3}{4} = 0$

9. $x^2 - 10x + y^2 + 6y - 47 = 0$

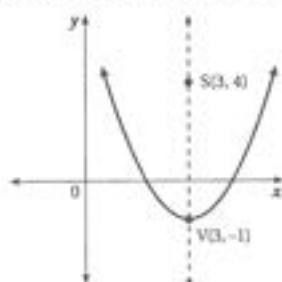
14. $x^2 + 8x + y^2 + 3y + 17\frac{1}{4} = 0$

Locus: the parabola and circle review

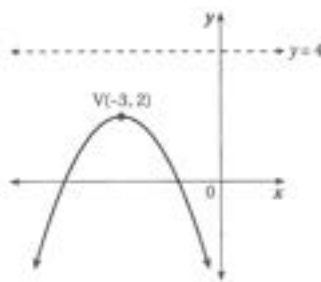
- Write down the equation of the locus of a point as it moves so that it is equidistant from the point $(0, 4)$ and the line $y = -4$.
- Write down the equation of the locus of a point as it moves so that it is equidistant from the point $(0, -3)$ and the line $y = 3$.
- A curve is drawn so that any point on the curve is equidistant from the point $(2, 4)$ and the line $y = 0$. What is the equation of the curve?
- Write down the equation of the locus of a point as it moves so that it is 3 units from the origin.
- Write down the equation of the locus of a point as it moves so that it is 2 units from the point $(-2, 1)$.
- A parabola has a vertex at $(0, 0)$ and a focus at $(0, 3)$. Write down the equation of the parabola.
- A parabola has a vertex at $(2, 1)$ and a focus at $(2, 3)$. What is the equation of the parabola?
- The equation of the directrix of a parabola is $y = 4$. The vertex is $(3, -1)$. Write down the equation of the parabola.
- Determine the equation of a parabola with a focus at $(2, -1)$ and directrix $y = 2$.
- A circle has a centre at $(-3, -3)$ and a radius of $\sqrt{7}$ units. Write down the equation of the circle in general form.
- Determine the equation of the directrix and the coordinates of the vertex and focus of a parabola with equation $(x - 3)^2 = 4(y + 5)$.
- Determine the equation of the directrix and the coordinates of the vertex and focus, given the equation of the parabola $x^2 - 2x - 4y - 3 = 0$.
- Draw a neat sketch of the parabola $x^2 = 8(y + 1)$, indicating the focus and the directrix.

Write down the equations of the following:

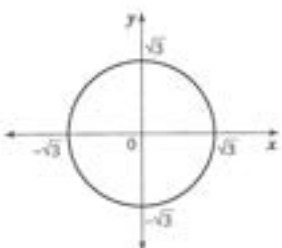
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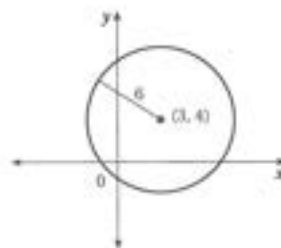
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16.



17.



- The axis of a parabola is parallel to the x -axis and has a vertex at $(2, -1)$. If the focal length is 3 units, write down a possible equation for the parabola.
- Does the point $(\sqrt{2}, \sqrt{2})$ lie on the circle, centre $(1, -1)$, radius $\sqrt{6}$ units?
- The point $(4, -2)$ lies on a circle, centre $(1, 2)$. Determine the equation of the circle.

1. Recognition of arithmetic sequences

- 1 12, 13 2 No 3 40, 50 4 2, 4 5 No 6 11, 15
 7 32, 40 8 18, 22 9 30, 39 10 No 11 8, 12 12 No
 13 1, 3 14 No 15 134, 145 16 3, 1 17 No
 18 -5, -9 19 -16, -18 20 -8, -4 21 $1, \frac{5}{4}$ 22 $\frac{2}{5}, \frac{1}{2}$
 23 $\frac{5}{12}, \frac{1}{2}$ 24 No 25 $4\frac{3}{4}, 5\frac{1}{2}$ 26 $\frac{3}{5}, \frac{3}{10}$ 27 0.2, 0.25
 28 5.6, 6.4 29 7.2, 6.8 30 No 31 $4a, 5a$
 32 $4a + 9, 5a + 12$ 33 No 34 No 35 $10\sqrt{3}, 13\sqrt{3}$
 36 No 37 $4\sqrt{6} + 10, 5\sqrt{6} + 12$ 38 $3 + \sqrt{27}, 3 + \sqrt{48}$
 39 $\sqrt{32}, \sqrt{50}$ 40 No

2. Finding terms of arithmetic sequences

- 1 18 2 29 3 44 4 42 5 96 6 60 7 123 8 23
 9 232 10 219 11 -32 12 -88 13 -82 14 52
 15 332 16 13.5 17 22.3 18 -7.8 19 -52
 20 -16.4 21 29 22 $6\frac{5}{8}$ 23 0.4 24 -1.4 25 1, 11
 26 -6.18 27 15.3 28 -10, -1 29 -4, -9 30 $-\frac{1}{4}, 0$
 31 6.2, 6.8 32 $-\frac{3}{5}, -\frac{1}{5}$

3. Recognising terms of arithmetic sequences

- 1 25th 2 30th 3 40th 4 27th 5 46th 6 36th
 7 31st 8 27th 9 21st 10 45th 11 43rd 12 48th
 13 19th 14 73rd 15 113th 16 16th 17 47th
 18 29th 19 14th 20 33rd 21 27th 22 41st
 23 120th 24 28th 25 45th 26 28th 27 29th
 28 45th 29 27th 30 35th

4. Finding the arithmetic sequence given two terms

- 1 $a = 6, d = 3$ 2 $a = -2, d = 2$ 3 $a = -6, d = 2$
 4 $a = 1, d = 7$ 5 $a = 9, d = 7$ 6 $a = -6, d = 5$
 7 $a = -3, d = -3$ 8 $a = 10, d = \frac{1}{2}$ 9 $a = \sqrt{3}, d = 3$
 10 $a = 6, d = \sqrt{3}$ 11 -336, -312 12 -1.5, -1.3
 13 -4.5, -12 14 2, -2 15 0.5, 1.8 16 $-2\frac{1}{4}, -\frac{1}{4}$
 17 $\frac{1}{5}, -\frac{1}{5}$ 18 $\frac{1}{5}, \frac{8}{15}$ 19 $-2\frac{1}{2}, -2\frac{9}{20}$ 20 $\sqrt{3}, -\sqrt{3}$

5. Finding a given term of an arithmetic sequence

- 1 38th 2 31st 3 No 4 35th 5 25th 6 No
 7 48th 8 28th 9 No 10 No 11 35th 12 29th
 13 51st 14 44th 15 No 16 No 17 40th 18 No
 19 39th 20 34th 21 $T_{31} = 188$ 22 $T_{61} = 117$
 23 $T_{414} = 2072$ 24 $T_{30} = 225$ 25 $T_{10} = -278$

6. The sum of an arithmetic series (1)

- 1 1070 2 450 3 1224 4 528 5 546 6 1905
 7 480 8 -690 9 -294 10 -551 11 -324
 12 -156 13 305 14 657 15 -162 16 -630
 17 $662\frac{1}{2}$ 18 $-42\frac{1}{2}$ 19 $189\frac{3}{4}$ 20 $-67\frac{1}{3}$ 21 10
 22 3 23 15 24 17 25 20

7. The sum of an arithmetic series (2)

- 1 1584 2 1155 3 -1935 4 1008 5 -750 6 1116
 7 1090 8 531 9 -595 10 100 11 $36\frac{1}{8}$ 12 -80
 13 $1600\sqrt{2}$ 14 212 15 97, 90, 83

8. Sigma notation and arithmetic series problems

- 1 110 2 196 3 480 4 108 5 847 6 -341
 7 765 8 1245 9 -189 10 $178\frac{1}{2}$ 11 6, 8, 10
 12 5, 9, 13 13 -4, 2, 8 14 -7, -3, 1 15 9, 3, -3

9. Arithmetic sequences and series review

- 1 No 2 Yes 3 -18, -9 4 146 5 2211 6 -4800
 7 12th 8 Yes 9 11, 19, 27, 35, 43 10 20 11 14554
 12 $T_{18} = -1$ 13 1120 14 $a = 8, d = 5$ 15 $a = 6b - 4c$
 16 4, 9, 14 17 184 18 5734 19 15.6 km 20 2340 km

10. Recognition of geometric sequences

- 1 16, 32 2 No 3 No 4 80, 160 5 No
 6 -128, -512 7 No 8 32, -64 9 -54, -162
 10 10, 5 11 243, 729 12 No 13 2.4, 4.8 14 No
 15 $1, \frac{1}{2}$ 16 No 17 $\frac{7}{8}, \frac{7}{16}$ 18 -40, -80 19 $-\frac{1}{16}, -\frac{1}{32}$
 20 10, 20 21 $4, 4\sqrt{2}$ 22 $8\sqrt{3}, 16\sqrt{3}$ 23 $\frac{1}{ab^{3n}}, \frac{1}{ab^4}$
 24 No 25 $\frac{a^3x}{y^4}, \frac{a^4x}{y^5}$ 26 No 27 3, 48 28 -32, -128
 29 $\frac{1}{5}, 125$ 30 -21, $-\frac{7}{27}$ 31 2, 1 32 32.4, 97.2
 33 -4.8, -8.2944 34 $4, \frac{1}{324}$ 35 -4.3, -0.86 36 $2, \frac{32}{81}$

11. Finding terms of geometric sequences

- 1 243 2 -384 3 31250 4 2048 5 -256 6 $\frac{5}{64}$
 7 $-\frac{1}{243}$ 8 128 9 $5461\frac{1}{3}$ 10 -4096 11 $-\frac{3}{32}$
 12 3.4×10^{-7} 13 $\frac{x}{729}$ 14 $10935x$ 15 $2x^{10}$
 16 $-\frac{3b}{64}$ 17 3 18 -2 19 6 20 $\frac{1}{3}$ 21 2 22 $-\frac{1}{9}$
 23 4 24 $\frac{3}{2}$ 25 $\frac{x^{13}}{64y^7}$ 26 $\frac{3}{\sqrt{2}}$

12. Recognising terms of geometric sequences

- 1 10th 2 9th 3 10th 4 15th 5 9th 6 7th
 7 9th 8 12th 9 13th 10 8th 11 11th 12 11th
 13 10th 14 11th 15 10th 16 14th 17 11th
 18 16th 19 6th 20 7th 21 7th 22 No 23 22nd
 24 No

13. Finding the geometric sequence given two terms

- 1 $a = 4, r = 2$ 2 $a = 2, r = 3$ 3 $a = 5, r = 3$
 4 $a = -4, r = 3$ 5 $a = -2, r = 7$ 6 $a = -0.7, r = 2$
 7 $a = -5, r = 2$ 8 $a = 10, r = 5$ 9 $a = 6, r = \frac{1}{2}$
 10 $a = -6, r = \frac{2}{3}$ 11 $a = 3, r = \frac{1}{4}$ 12 $a = 0.2, r = 3$
 13 $a = -9, r = \frac{1}{3}$ 14 $a = -90, r = -\frac{1}{3}$ 15 $a = 6, r = 0.3$
 16 $a = 20, r = 1.5$

14. The sum of a geometric series

- 1 349 525 2 29 523 3 19 682 4 889 5 -1530
 6 -4092 7 43 690 8 21 845 9 $29 524\frac{1}{3}$ 10 $-23\frac{81}{64}$
 11 129.16 12 6.25 13 $341\frac{1}{4}$ 14 $\frac{a(b^{10} - 1)}{b - 1}$
 15 $\frac{x^{12} - 1}{x(x - 1)}$ 16 $31\sqrt{2} + 30$ 17 765 18 4.00
 19 $\frac{85}{128}$ 20 5.00 21 -1022 22 125.00

15. The limiting sum of a geometric series

- 1 20 2 No 3 200 4 No 5 $1\frac{1}{2}$ 6 No 7 No
 8 $31\frac{1}{4}$ 9 18 10 No 11 $21\frac{1}{3}$ 12 $2\frac{2}{3}$ 13 $\frac{1}{2}$ 14 $\frac{4}{7}$
 15 $-5\frac{1}{3}$ 16 $2\frac{1}{2}$ 17 7.5 18 10.5 19 $12\frac{1}{2}$ 20 32
 21 $a = 7$ 22 24 23 2000 m

16. Recurring decimals and the limiting sum

- 1 $\frac{2}{9}$ 2 $\frac{4}{9}$ 3 $\frac{7}{33}$ 4 $\frac{5}{33}$ 5 $\frac{8}{99}$ 6 $\frac{17}{99}$ 7 $\frac{29}{99}$ 8 $\frac{7}{90}$
 9 $\frac{2}{45}$ 10 $\frac{35}{99}$ 11 $\frac{17}{99}$ 12 $\frac{4}{45}$ 13 $\frac{6}{9}$ 14 $5\frac{14}{99}$
 15 $6\frac{13}{33}$ 16 $2\frac{31}{99}$ 17 $\frac{13}{30}$ 18 $\frac{5}{18}$ 19 $\frac{49}{99}$ 20 $\frac{293}{990}$
 21 $6\frac{143}{300}$ 22 $2\frac{37}{300}$

17. Compound interest

- 1 \$2533.54 2 \$8539.87 3 \$1229.26 4 \$4343.01
 5 \$3262.04 6 \$3869.68 7 \$100 550.96 8 \$89.01
 9 \$2332.42 10 \$18 524.16 11 \$12 570.99
 12 \$11 111.67 13 \$2365.60 14 \$5983.40 15 \$9890.82

18. Superannuation

- 1 \$82 697.91 2 \$109 614.30 3 \$87 789.33 4 \$50 457.60
 5 \$58 454.56 6 Monthly, by \$744.33
 7 First alternative, by \$58 008.14

19. Geometric sequences and series review

- 1 Yes 2 No 3 81, 243 4 $2\frac{14}{25}$ 5 13 122 6 3
 7 $\frac{7}{11}$ 8 8191 $\frac{3}{4}$ 9 31.25 10 2, 6, 18, 54, 162 11 24
 12 6 13 $a = 4, r = \frac{1}{2}$ 14 $b = \sqrt{ac}$ 15 10.67
 16 $\frac{3}{2}, 3, 6, r = 2$ 17 $\frac{1}{3}$

20. Using the general expressions S_n and T_n in sequences and series

- 1 5, 8, 11 2 -1, -5, -9 3 2, 8, 32 4 330
 5 Arithmetic, $a = 6, d = 2, T_{21} = 46$ 6 2592, 93 312
 7 $\frac{3}{9}, 3$ 8 $T_n = 4n - 7$ 9 $n = 51, 51\text{st term}$
 10 $n = 5, 5\text{th term}$ 11 $11 - 4 \neq 30 - 11$ 12 87
 13 $-11\frac{29}{32}$ 14 910 15 $S_1 = 4, S_2 = 11, a = 4, d = 3, T_7 = 22$
 16 16 17 $a = 64, r = \frac{1}{2}$

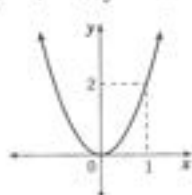
21. Sequences and series review

- 1 80 2 576 3 29 524 4 $\frac{1}{4096}$ 5 78 125 6 $1\frac{511}{512}$
 7 $\frac{1}{512}$ 8 8 9 -4, -32 10 6, 14 11 27, $\frac{1}{3}$
 12 -6, -486 13 $\frac{71}{99}$ 14 Yes, 14th 15 11th 16 12
 17 86, 97, 108 18 3, 5, 7, $S_{15} = 255$ 19 $3\frac{63}{64}$ 20 $6\frac{1}{4}$
 21 29th 22 3, 12, 48 23 No, $T_2 - T_1 \neq T_3 - T_2$ 24 $171\sqrt{x}$

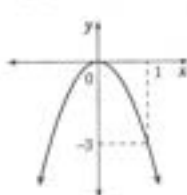
22. Sketching quadratics (basic parabolas)

- 1 $y = 4$ 2 $y = -3$ 3 $x = 4, y = -16$ 4 $x = \pm\sqrt{7}, y = -7$

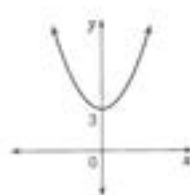
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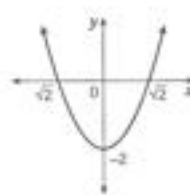
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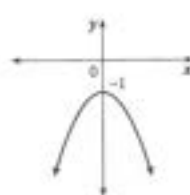
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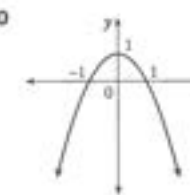
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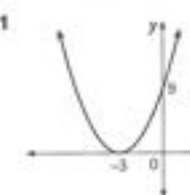
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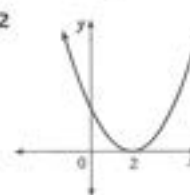
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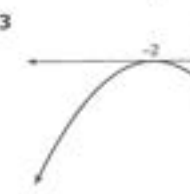
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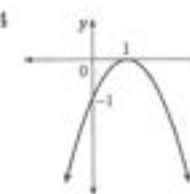
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13



14



- 15 A 16 D 17 B 18 F 19 C 20 E

- 21 $y = -x^2 + 9$ 22 $y = x^2 - 1$ 23 $y = (x - 1)^2$
 24 $y = -(x + 2)^2$ 25 $y = -x^2 + 5$ 26 $y = x^2 + 4$
 27 $y = -x^2 - 4$ 28 $y = x^2 + 3$ 29 $y = -(x - 2)^2$
 30 $y = (x + 3)^2$

23. Sketching parabolas using the intercepts and vertex

- 1 $x = -3, -1, y = 3, V(-2, -1)$ 2 $x = 1, 5, y = 5, V(3, -4)$
 3 $x = 0, 2, y = 0, V(1, -1)$ 4 $x = 0, 3, y = 0, V(1\frac{1}{2}, 2\frac{1}{4})$
 5 $x = -3, -2, y = 6, V(-2\frac{1}{2}, -\frac{1}{4})$
 6 $x = -6, 2, y = -12, V(-2, -16)$
 7 $x = -3, 1, y = 3, V(-1, 4)$
 8 $x = -2, -5, y = -10, V(-3\frac{1}{2}, 2\frac{1}{4})$ 9 $x = -4, 3, y = -12$
 10 $x = -2, 5, y = -10$ 11 $x = -6, 1, y = -6$
 12 $x = \frac{1}{2}, 3, y = 3$ 13 $x = -\frac{2}{3}, -5, y = 10$
 14 $x = -\frac{3}{2}, -2, y = 6$ 15 $x = 1\frac{1}{2}, -\frac{1}{2}, y = -3$
 16 $x = 1\frac{1}{2}, 5, y = 15$ 17 $x = -4, 2, y = 8$
 18 $x = -1, 3, y = 3$ 19 $x = -3, 4, y = 12$
 20 $x = \frac{1}{2}, 2, y = -2$

24. Solving quadratic equations using the formula

- 1 $\frac{-3 \pm \sqrt{5}}{2}$ 2 $\frac{3 \pm \sqrt{29}}{2}$ 3 $\frac{-7 \pm \sqrt{73}}{4}$ 4 $2 \pm \sqrt{5}$
 5 $-2 \pm \sqrt{10}$ 6 $3 \pm \sqrt{14}$ 7 $-2, -\frac{1}{2}$ 8 $\frac{2}{3}, \frac{1}{2}$
 9 $\frac{-5 \pm \sqrt{33}}{4}$ 10 $\frac{3 \pm \sqrt{41}}{4}$ 11 -2.30, 1.30
 12 -2.82, -0.18 13 -0.59, -3.41 14 -2.64, 1.14
 15 -0.59, 0.84 16 2.85, -0.35 17 $1\frac{2}{3}, 1$ 18 -0.63, -6.37
 19 -0.44, 3.44 20 2.31, 0.09

25. The discriminant of a quadratic equation

- 1 real, equal, rational 2 unreal 3 real, unequal, irrational
 4 real, unequal, rational 5 real, equal, rational 6 unreal
 7 real, equal, rational 8 unreal 9 real, unequal, rational
 10 unreal 11 real, equal, rational
 12 real, unequal, rational 13 $k = \frac{25}{8}$ 14 $k > \frac{9}{8}$
 15 $k < \frac{49}{4}$ 16 $k = 4$ 17 $k < 2$
 18 $\Delta = 729$, which is a perfect square, $\therefore$ rational roots

26. Sums and products of roots of quadratic equations

- 1 2 2 $\frac{3}{2}$ 3 4 4 1 5 24 6 -2 7 72 8 80
 9 $8\frac{1}{2}$ 10 $2\frac{1}{3}$ 11 $x^2 - 4x + 3 = 0$ 12 $x^2 + 4x + 8 = 0$
 13 $5x^2 - x - 25 = 0$ 14 $12x^2 - 4x + 3 = 0$
 15 $6x^2 + 7x - 3 = 0$ 16 $x^2 - 4x - 12 = 0$

27. The discriminant and the parabola

- 1 $\Delta > 0, a < 0$ 2 $\Delta = 0, a < 0$ 3 $\Delta > 0, a > 0$
 4 $\Delta < 0, a > 0$ 5 $\Delta < 0, a < 0$ 6 $\Delta = 0, a > 0$
 7 $\Delta = 0, a < 0$ 8 $\Delta > 0, a > 0$ 9 $\Delta > 0, a < 0$
 10 $\Delta < 0, a < 0$ 11 $\Delta = 0, a < 0$ 12 $\Delta = 0, a > 0$
 13 $\Delta > 0, a < 0$ 14 $\Delta = 0, a > 0$ 15 B 16 A 17 A
 18 A 19 Positive definite as $a > 0$, Δ is negative
 20 $\Delta = 0$, $\therefore$ equal roots, parabola touches the x -axis, $a > 0$,
 concave up $\therefore$ positive definite
 21 $\Delta > 0$, $\therefore$ parabola cuts x -axis twice 22 $k > \frac{1}{8}$
 23 $m < \frac{9}{16}$ 24 $a > \frac{1}{3}$ 25 $k > 2$ 26 $p = 3$

28. Roots of quadratic equations review

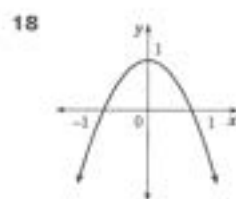
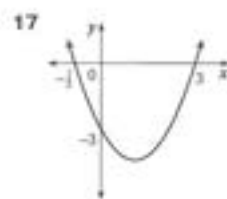
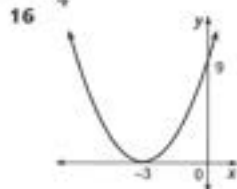
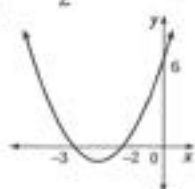
- 1 $\Delta = 0$, $\therefore$ equal roots 2 $\Delta = -7$, $\therefore$ no real roots
 3 $\Delta = 16$, $\therefore$ 2 real roots 4 $\Delta = -11$, $\therefore$ no roots 5 5
 6 3 7 19 8 13 9 -3 10 $\frac{2}{3}$ 11 $x^2 - 6x - 4 = 0$
 12 $6x^2 + 9x + 4 = 0$ 13 $x^2 - 12x + 32 = 0$ 14 $k = \frac{9}{4}$
 15 $k = 7$ 16 4 17 9 18 42 19 19 20 $-\frac{4}{9}$ 21 3

29. Solving quadratic equations by completing the square

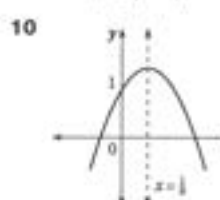
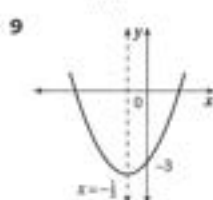
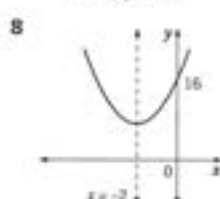
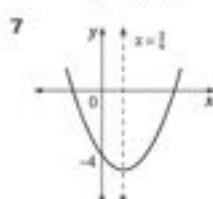
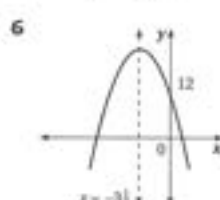
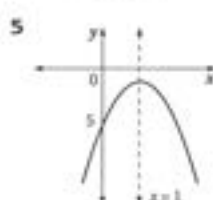
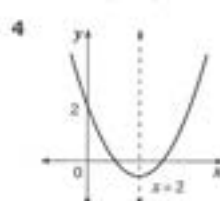
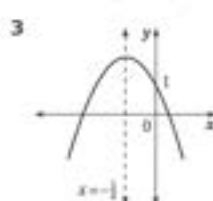
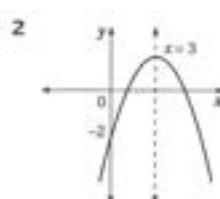
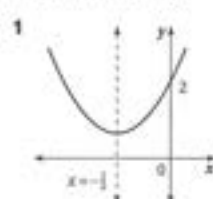
- 1 4 2 6, 3 3 81, 9 4 24, 36 5 $3, \frac{9}{4}, x, \frac{3}{2}, \pm \frac{\sqrt{11}}{2}$
 6 $-2 \pm \sqrt{5}$ 7 $-3 \pm \sqrt{7}$ 8 $\frac{3 \pm \sqrt{5}}{2}$ 9 $\frac{5 \pm \sqrt{17}}{2}$
 10 $\frac{-7 \pm \sqrt{65}}{2}$ 11 $-2 \pm \sqrt{11}$ 12 $\frac{-4 \pm \sqrt{2}}{2}$
 13 $\frac{-9 \pm \sqrt{65}}{4}$ 14 $\frac{-3 \pm \sqrt{57}}{6}$ 15 $\frac{5 \pm \sqrt{55}}{5}$

30. Quadratic equations and functions review

- 1 ± 5 2 $\pm \sqrt{6}$ 3 0, -4 4 0, -3 5 4, -3 6 -2, 3
 7 -4 8 5 9 $-\frac{3}{2}, -4$ 10 $-\frac{4}{3}, -\frac{3}{2}$ 11 $\frac{-1 \pm \sqrt{13}}{2}$
 12 $\frac{-7 \pm \sqrt{29}}{2}$ 13 $-\frac{5}{2}, 1$ 14 $\frac{9 \pm \sqrt{57}}{4}$
 15



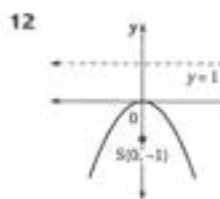
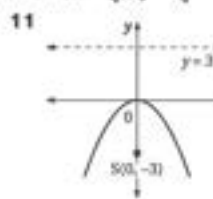
- 19 $x = -2, -\frac{1}{2}, y = 2$ 20 $(-3, -8)$ 21 $y = x^2 - 8x + 15$
 22 $y = -2x^2 - 3x + 5$

31. Graphing quadratics using the discriminant

- 11 A 12 B 13 C 14 E 15 G 16 I

32. The parabola as a locus: vertex the origin

- 1 $S(0, 1), y = -1$ 2 $S(0, 2), y = -2$ 3 $S(0, 4), y = -4$
 4 $S(0, 3), y = -3$ 5 $S(0, \frac{1}{4}), y = -\frac{1}{4}$ 6 $S(0, -2), y = 2$
 7 $S(0, -\frac{3}{4}), y = \frac{3}{4}$ 8 $S(0, -\frac{1}{2}), y = \frac{1}{2}$ 9 $S(0, -1), y = 1$
 10 $S(0, -\frac{5}{4}), y = \frac{5}{4}$



- 13 $x^2 = 14y$ 14 $x^2 = 20y$ 15 $x^2 = 28y$ 16 $x^2 = 10y$

17 $x^2 = -24y$ 18 $x^2 = -\frac{3}{2}y$ 19 $x^2 = -\frac{1}{2}y$ 20 $x^2 = -y$
 21 $x^2 = -9y$

33. The parabola as a locus: vertex not the origin (1)

1 $V(3, 1)$ $S(3, 2)$ $y = 0$ 2 $V(2, 3)$ $S(2, 6)$ $y = 0$
 3 $V(5, -3)$ $S(5, 1)$ $y = -7$ 4 $V(-3, 2)$ $S(-3, 3)$ $y = 1$
 5 $V(-2, -3)$ $S(-2, -5)$ $y = -1$ 6 $V(-4, -1)$ $S(-4, -4)$ $y = 2$
 7 $V(1, -3)$ $S(1, -2\frac{1}{2})$ $y = -3\frac{1}{2}$ 8 $V(-1, 1)$ $S(-1, -\frac{1}{2})$ $y = 2\frac{1}{2}$
 9 $V(0, -4)$ $S(0, -2)$ $y = -6$ 10 $V(3, -1)$ $S(3, -1\frac{1}{4})$ $y = -\frac{3}{4}$
 11 $V(2, \frac{1}{2})$ $S(2, 1)$ $y = 0$ 12 $V(2, 0)$ $S(2, -\frac{1}{2})$ $y = \frac{1}{2}$
 13 $V(3, 4)$ $S(3, 5)$ $y = 3$ 14 $V(-1, \frac{1}{2})$ $S(-1, -1\frac{1}{2})$ $y = 2\frac{1}{2}$
 15 $(x-3)^2 = 8(y-2)$ 16 $(x-2)^2 = 8(y+1)$
 17 $(x-3)^2 = -12(y-1)$ 18 $(x+3)^2 = -8(y+3)$
 19 $(x-1)^2 = 12(y+1)$ 20 $(x+1)^2 = -16(y-2)$

34. The parabola as a locus: vertex not the origin (2)

1 $(x-1)^2 = 4(y-2)$ 2 $(x+1)^2 = 8(y-3)$
 3 $(x+3)^2 = -12(y-2)$ 4 $(x+2)^2 = -8(y+1)$
 5 $(x+2)^2 = 12y$ 6 $(x-4)^2 = -12(y-2)$
 7 $(x+2)^2 = 16(y+3)$ 8 $(x-3)^2 = 8(y-4)$
 9 $(x-2)^2 = -12(y-1)$ 10 $(x-2)^2 = 12y$
 11 $(x-3)^2 = 24(y-4)$ 12 $(x+2)^2 = 6(y+\frac{1}{2})$
 13 $(x-2)^2 = -12(y+1)$ 14 $(x-4)^2 = 8(y-3)$
 15 $(x-3)^2 = \pm 20(y+1)$ 16 $(x-3)^2 = -12(y-4)$

35. The circle as a locus

1 $(0, 0)$, $r = 2$ 2 $(0, 0)$, $r = 3$ 3 $(0, 0)$, $r = 1$
 4 $(0, 0)$, $r = \sqrt{3}$ 5 $(3, 4)$, $r = 4$ 6 $(-1, -2)$, $r = 2$
 7 $(-2, -3)$, $r = 5$ 8 $(2, -3)$, $r = 2\sqrt{2}$ 9 $(2, -1)$, $r = \frac{1}{2}$
 10 $(-3, 2)$, $r = \frac{2}{3}$ 11 $x^2 + y^2 = 25$ 12 $x^2 + y^2 = 5$
 13 $(x-3)^2 + (y-3)^2 = 9$ 14 $(x-4)^2 + (y-5)^2 = 4$

15 $(x+3)^2 + (y+2)^2 = 6\frac{1}{4}$ 16 $(x-4)^2 + (y+5)^2 = 7$
 17 $x^2 + y^2 = 25$ 18 $x^2 + y^2 = 13$
 19 $(x-3)^2 + (y-4)^2 = 18$ 20 Yes 21 Outside

36. Finding the focus, vertex and directrix by completing the square

1 $V(2, 3)$ $S(2, 5)$, $y = 1$ 2 $V(3, -1)$ $S(3, 0)$, $y = -2$
 3 $V(8, 12)$ $S(8, 11)$, $y = 13$ 4 $V(3, -6)$ $S(3, -5\frac{1}{2})$, $y = -6\frac{1}{2}$
 5 $V(-2, -\frac{1}{2})$ $S(-2, 1\frac{1}{2})$, $y = -2\frac{1}{2}$ 6 $V(7, 10)$ $S(7, 9)$, $y = 11$
 7 $V(1, -\frac{1}{2})$ $S(1, \frac{1}{2})$, $y = -1\frac{1}{2}$ 8 $V(-3, \frac{1}{2})$ $S(-3, -1\frac{1}{2})$, $y = 2\frac{1}{2}$
 9 $V(-5, -1)$ $S(-5, -2\frac{1}{2})$, $y = \frac{1}{2}$
 10 $V(-4, -2)$ $S(-4, -1\frac{3}{4})$, $y = -2\frac{1}{4}$
 11 $y = \frac{1}{16}(x^2 - 4x + 20)$ 12 $y = \frac{1}{4}(x^2 - 6x + 5)$
 13 $y = -\frac{1}{8}(x^2 - 2x - 7)$ 14 $y = -\frac{1}{12}(x^2 + 4x - 20)$

37. Equations of circles in expanded form

1 $x^2 + y^2 - 6x - 8y + 9 = 0$ 2 $x^2 + y^2 + 10x - 4y + 28 = 0$
 3 $x^2 + y^2 + 6y + 6 = 0$ 4 $x^2 + y^2 + 4x - 8y + 15 = 0$
 5 $r = 6$, $(3, 2)$ 6 $r = 1$, $(2, -1)$ 7 $r = 5$, $(-5, 4)$
 8 $r = 4$, $(-1, -4)$ 9 $r = 9$, $(5, -3)$ 10 $r = 3$, $(0, -3)$
 11 $r = 5$, $(-3, 1)$ 12 $r = 1$, $(1, 2)$ 13 $r = 4$, $(\frac{1}{2}, 1)$
 14 $r = 1$, $(-4, -1\frac{1}{2})$

38. Locus: the parabola and circle review

1 $x^2 = 16y$ 2 $x^2 = -12y$ 3 $(x-2)^2 = 8(y-2)$
 4 $x^2 + y^2 = 9$ 5 $(x+2)^2 + (y-1)^2 = 4$ 6 $x^2 = 12y$
 7 $(x-2)^2 = 8(y-1)$ 8 $(x-3)^2 = -20(y+1)$
 9 $(x-2)^2 = -6(y-\frac{1}{2})$ 10 $x^2 + y^2 + 6x + 6y + 11 = 0$
 11 $y = -6$, $V(3, -5)$ $S(3, -4)$ 12 $y = -2$, $V(1, -1)$ $S(1, 0)$
 13 $S(0, 1)$, $y = -3$ 14 $(x-3)^2 = 20(y+1)$
 15 $(x+3)^2 = -8(y-2)$ 16 $x^2 + y^2 = 3$
 17 $(x-3)^2 + (y-4)^2 = 36$ 18 $(x-2)^2 = \pm 12(y+1)$
 19 Yes 20 $(x-1)^2 + (y-2)^2 = 25$

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